Software Requirements Specification
for
docWise

Version 1.1 approved

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docwise Organization

4.7.2025

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1. Introduction

1.1. Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the functional and non-functional requirements for the docWise system - an AI-powered medical diagnosis and decision support platform. The document is intended for:

- Software developers and architects responsible for implementing the system
- Project managers overseeing the development process
- Quality assurance teams conducting testing and validation
- Stakeholders including healthcare professionals, hospital administrators, and regulatory bodies
- Maintenance and support teams for future system updates

This SRS serves as the foundation for system design, implementation, testing, and acceptance criteria, ensuring all stakeholders have a clear understanding of the system's capabilities and constraints.

1.2. Scope

The docWise system is designed to address critical issues in healthcare diagnostics, specifically targeting the reduction of medical misdiagnoses and elimination of diagnostic discrimination. The system encompasses:

In Scope:

- AI-assisted diagnostic recommendations for healthcare professionals
- Secure user authentication through e-Government (e-Devlet) integration
- Integration with Turkey's national health platform (e-Nabız) for automatic health data transfer
- Patient health data analysis and graphical reporting
- AI-based lifestyle recommendations for patients
- Treatment plan management and validation
- Administrative oversight and compliance monitoring
- Ministry of Health inspection board integration
- Hospital management functions for both public and private healthcare facilities

- Laboratory test result management and sharing

Target Users:

- Patients seeking medical diagnosis and health monitoring
- Medical doctors requiring diagnostic decision support
- Hospital administrators managing patient care workflows
- Ministry of Health inspection board officials
- Laboratory staff processing medical tests
- System administrators overseeing platform operations

Primary Benefits:

- Reduction of misdiagnosis rates by more than 50%
- Elimination of gender and ethnicity-based diagnostic discrimination
- Enhanced patient safety through prevention of unnecessary treatments
- Improved diagnostic accuracy with AI accuracy target of 75% or higher
- Streamlined healthcare workflows and data management

1.3. References

1. **Vision and Scope Document for docWise**, Version 1.1, Rukiye Sedef Keskin, Docwise Organization, 04.05.2025
2. **Use Case Document for docWise**, Rukiye Sedef Keskin, Docwise Organization, 2025
3. **Turkey Personal Data Protection Law (KVKK) -** Legal framework for data privacy compliance
4. **Health Insurance Portability and Accountability Act (HIPAA)** - Healthcare data protection standards
5. **General Data Protection Regulation (GDPR)** - European Union data protection regulation
6. **e-Government Integration Standards** - Turkish e-Devlet platform documentation
7. **e-Nabız Integration Specifications** - National health system integration requirements

8. **IEEE Standard 830-1998 -** IEEE Recommended Practice for Software Requirements Specifications

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2. Overall Description

2.1. Vision Statement

The docWise system operates as a comprehensive AI-powered medical diagnosis and decision support platform that integrates with Turkey's national healthcare infrastructure. The system serves as an intermediary between healthcare providers, patients, and regulatory bodies, facilitating transparent and accountable medical decision-making processes.

2.1.1 External Interfaces

The docWise system interfaces with several external systems to provide comprehensive healthcare services:

e-Government (e-Devlet) Interface:

- Secure user authentication and identity verification
- Real-time credential validation
- User registration and profile management
- Bi-directional data exchange using secure protocols

e-Nabız National Health Platform Interface:

- Automatic health data import and synchronization
- Medical history retrieval
- Test result integration
- Compliance reporting to national health authorities

2.1.2 User Interfaces

The system provides intuitive web-based interfaces optimized for different user roles:

Input Methods:

- Users will interact with the system using standard computer peripherals including mouse, touchpad, and keyboard
- The "Tab" key will be available to navigate between screen elements for accessibility
- Touch screen support for mobile and tablet devices

Interface Characteristics:
- Responsive web design supporting desktop, tablet, and mobile devices
- Multi-language support (Turkish and English)
- Dark mode and light mode themes for user preference

Role-Specific Interfaces:
- **Patient Portal:** Simplified interface for viewing health data, diagnoses, and lifestyle recommendations
- **Doctor Dashboard:** Clinical interface with diagnostic tools, patient management, and AI assistance
- **Admin Panel:** Administrative interface for oversight, reporting, and system management
- **Laboratory Interface:** Streamlined workflow for test result entry and management

2.1.3 Hardware Interfaces

Client-Side Requirements:
- Standard desktop computers, laptops, tablets, or smartphones
- Minimum screen resolution: 1024x768 pixels (desktop), 375x667 pixels (mobile)
- Internet connectivity: Minimum 1 Mbps for basic functionality, 5 Mbps recommended
- Input devices: Keyboard, mouse/touchpad, or touch screen

Server-Side Infrastructure:
- High-performance servers with multi-core processors (minimum 16 cores)
- Minimum 64GB RAM for AI processing and database operations
- SSD storage with minimum 10TB capacity for data storage
- Hardware security modules (HSM) for cryptographic operations

2.1.4 Software Interfaces

Required Client Software:
- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)
- JavaScript enabled for dynamic functionality
- PDF reader for document download and viewing
- No additional software installation required on client side

Server Software Stack:
- Operating System: Linux-based server (Ubuntu 20.04 LTS or CentOS 8)

- Web Server: Apache HTTP Server or Nginx
- Application Server: Node.js or JAVA-based framework
- Database: PostgreSQL 13+ for document storage
- AI/ML Framework: TensorFlow or PyTorch for machine learning operations
- Security: SSL/TLS encryption, OAuth 2.0 authentication

Integration Software:

- e-Devlet API integration libraries
- e-Nabız platform connectors
- HL7 FHIR compliance libraries for healthcare data exchange
- PDF generation libraries for report creation

2.1.5 Communication Interfaces

Primary Communication Protocols:

- **HTTPS/TLS 1.3:** Secure web communication for all user interactions
- **REST APIs:** For internal system communication and external integrations
- **SOAP:** For legacy healthcare system integration where required
- **WebSocket:** For real-time notifications and updates

Specific Interface Requirements:

- **e-Devlet Integration:** HTTPS with mutual TLS authentication
- **e-Nabız Platform:** Secure API calls using government-approved protocols
- **AI Model Communication:** Internal high-speed TCP connections for model inference
- **Database Communication:** Encrypted database connections using TLS
- **Email Notifications:** SMTP over TLS for system notifications

Data Formats:

- JSON for API communications
- HL7 FHIR for healthcare data exchange
- XML for government system integration
- PDF for document generation and sharing

2.1.6 Oprations

The docWise system operates in a single primary mode with multiple sub-operational states:

Operational Mode: The system maintains continuous 24/7 operation with the following states:

- **Active State:** Full system functionality available to all users
- **Maintenance State:** Limited functionality during system updates (scheduled during low-usage periods)
- **Emergency State:** Core diagnostic functions only during system issues
- **Backup State:** Automatic failover to backup systems during primary system failure

Operational Characteristics:

- **Availability Target:** 99.9% uptime (maximum 8.76 hours downtime per year)
- **Maintenance Windows:** Scheduled weekly maintenance (Sunday 2:00-4:00 AM local time)
- **Disaster Recovery:** Automatic backup systems with Recovery Time Objective (RTO) of 4 hours
- **Load Balancing:** Automatic traffic distribution across multiple servers
- **Monitoring:** Continuous system health monitoring with automated alerting

User Session Management:

- Maximum session duration: 8 hours for security
- Automatic session extension for active users
- Immediate session termination for suspicious activity
- Session data encryption and secure storage

2.1.7 Site Adaption Requirements

Geographic Adaptations:

- **Turkey-Specific Requirements:** Full compliance with Turkish healthcare regulations (KVKK, MOH guidelines)
- **Language Localization:** Primary Turkish interface with English secondary support
- **Date/Time Formats:** Turkish locale formatting (DD.MM.YYYY, 24-hour time)
- **Currency Display:** Turkish Lira (TRY) for private healthcare pricing

Healthcare System Integration:

- **Public Hospitals:** Integration with Turkish public health infrastructure
- **Private Healthcare:** Support for insurance-based pricing and billing

Regulatory Compliance Adaptations:

- **Data Sovereignty:** All patient data stored within Turkish borders

- **Audit Trail Requirements:** Comprehensive logging per Turkish healthcare audit standards
- **Privacy Controls:** KVKK-compliant data protection measures
- **Government Reporting:** Automated compliance reporting to Turkish health authorities

2.2. Product Functions

The docWise system provides comprehensive healthcare diagnostic support through the following main functional areas:

2.2.1 User Authentication and Management

- **Secure Registration:** e-Government (e-Devlet) based user registration for patients and doctors
- **Multi-Role Authentication:** Role-based login system supporting patients, doctors, administrators, laboratory staff, and inspection board officers
- **Profile Management:** User profile updates including contact information, preferences
- **Session Management:** Secure session handling with automatic timeout and activity monitoring

2.2.2 AI-Powered Diagnostic Support

- **Intelligent Diagnosis Generation:** AI analysis of patient symptoms, test results, and medical history to generate diagnostic suggestions
- **Diagnostic Accuracy Tracking:** Continuous monitoring of AI diagnostic accuracy with target of 75% or higher
- **Machine Learning Enhancement:** Continuous improvement of AI models based on doctor feedback and new medical data
- **Multi-Parameter Analysis:** Integration of symptoms, laboratory results, imaging data, and patient history for comprehensive diagnosis

2.2.3 Patient Health Management

- **e-Nabız Integration:** Automatic import and synchronization of patient health data from Turkey's national health platform

- **Health History Tracking:** Comprehensive storage and display of patient medical history, diagnoses, and treatment plans
- **Graphical Health Analysis:** Visual representation of patient health trends and historical data
- **Lifestyle Recommendations:** AI-generated personalized health and lifestyle suggestions based on patient data
- **Document Management:** PDF generation and download of medical reports, diagnoses, and treatment plans

2.2.4 Clinical Workflow Management

- **Patient Queue Management:** Systematic handling of patient appointments and examination scheduling
- **Symptom Documentation:** Structured entry and storage of patient symptoms and clinical observations
- **Treatment Plan Creation:** Comprehensive treatment planning tools with validation and approval workflows
- **Test Request Management:** Laboratory test ordering, tracking, and result integration
- **Diagnosis Validation:** Doctor review, approval, or rejection of AI-generated diagnostic suggestions with mandatory justification

2.2.5 Hospital Administration Functions

- **Appointment Management:** Patient registration and scheduling for both public and private healthcare facilities
- **Insurance Processing:** Integration with patient insurance policies for pricing and coverage determination
- **Laboratory Coordination:** Test request processing, result upload, and data sharing with e-Nabız system
- **Billing and Pricing:** Automated pricing calculation based on insurance coverage and standard fee schedules
- **Resource Management:** Hospital workflow optimization and administrative oversight

2.2.6 Quality Assurance and Compliance

- **Administrative Oversight:** Comprehensive monitoring of doctor performance and treatment plan analysis
- **Necessity Analysis:** Evaluation of treatment plan appropriateness and medical necessity

- **Malicious Practice Detection:** Automated identification of potentially unnecessary or inappropriate treatments
- **Investigation Management:** Systematic tracking and management of doctors under investigation
- **Regulatory Reporting:** Automated compliance reporting to Ministry of Health inspection board

2.2.7 Data Integration and Reporting

- **Multi-Platform Data Sync:** Seamless data exchange between docWise, e-Nabız, and e-Government systems
- **Statistical Analysis:** Comprehensive reporting on diagnostic accuracy, treatment outcomes, and system performance
- **Audit Trail Management:** Complete logging of all system activities for compliance and security purposes
- **Performance Analytics:** Real-time monitoring of system usage, user satisfaction, and clinical outcomes

2.2.8 Security and Privacy Protection

- **Data Encryption:** End-to-end encryption of all patient data and communications
- **Access Control:** Role-based permissions ensuring users only access authorized information
- **Privacy Compliance:** Full adherence to KVKK, HIPAA, and GDPR regulations

Secure Communication: Encrypted data transmission for all external system integrations

2.3. User Characteristics

The docWise system is designed to accommodate diverse user groups with varying technical expertise and healthcare knowledge. Each user category has specific characteristics and requirements:

2.3.1 Patients

Technical Proficiency Requirements:

- Basic computer literacy with ability to navigate web browsers
- Familiarity with internet usage and online form completion
- Ability to use standard input devices (mouse, keyboard, or touchscreen)

System Access Requirements:

- Valid Turkish citizenship with e-Government (e-Devlet) account access

- Active e-Nabız account for health data integration

- Device with internet connectivity (computer, tablet, or smartphone)
- Modern web browser (Chrome 90+, Firefox 88+, Safari 14+, Edge 90+)

Accessibility Considerations:

- Support for users with visual impairments through high contrast modes
- Keyboard navigation for users with mobility limitations
- Multi-language support (Turkish primary, English secondary)

2.3.2 Medical Doctors

Professional Requirements:

- Licensed medical practitioner registered in Turkey
- Valid medical license and e-Government doctor credentials

Technical Proficiency Requirements:

- Intermediate computer skills with experience using electronic health records
- Proficiency in web-based applications and medical software systems
- Ability to navigate complex interfaces with multiple data inputs
- Understanding of digital medical data formats

System Access Requirements:

- e-Government doctor authentication credentials
- Professional workstation or laptop with stable internet connection
- Familiarity with clinical workflow management systems
- Ability to integrate AI recommendations with clinical judgment

2.3.3 Hospital Administrators

Professional Background:

- Healthcare administration experience
- Understanding of hospital operations and patient flow management
- Knowledge of insurance processing and healthcare billing
- Familiarity with regulatory compliance requirements

Technical Requirements:

- Proficiency in data analysis and reporting tools
- Experience with healthcare information systems

2.3.4 Laboratory Staff

Professional Requirements:

- Medical laboratory technician certification or equivalent
- Experience with laboratory information systems
- Understanding of medical test procedures and result interpretation
- Knowledge of laboratory quality control processes

Technical Proficiency:

- Familiarity with laboratory equipment interfaces
- Experience with test result entry
- Basic computer skills for data input
- Understanding of digital file formats for test results

System Integration:

- Understanding of test request processing workflows
- Knowledge of laboratory-to-hospital data sharing protocols

2.3.5 Ministry of Health Inspection Board Officers

Professional Authority:

- Official government position with healthcare oversight responsibilities
- Legal authority to investigate medical practice issues
- Understanding of healthcare regulations and compliance requirements
- Experience with official documentation and case management

Technical Requirements:

- Government-level security clearance and system access
- Advanced analytical skills for case review and evaluation
- Proficiency in official reporting and documentation systems
- Understanding of legal and regulatory documentation requirements

System Access:

- Highest level security credentials with audit and enforcement capabilities
- Ability to access comprehensive case files and historical data
- Authority to suspend user access and initiate investigations
- Responsibility for regulatory compliance monitoring

2.3.6 General System Requirements for All Users

Minimum Technical Environment:

- Stable internet connection (minimum 1 Mbps, recommended 5 Mbps)

- Modern operating system (Windows 10, macOS 10.15, Linux Ubuntu 18.04, or equivalent)
- Updated web browser with JavaScript enabled
- PDF reader software for document viewing

Security and Compliance Awareness:

- Understanding of personal data protection principles
- Compliance with organizational security policies
- Awareness of appropriate system usage guidelines
- Knowledge of incident reporting procedures for security concerns

2.4. Constraints

The docWise system operates within various technical, regulatory, and operational constraints that influence system design and implementation decisions.

2.4.1 Regulatory Policies

Turkish Healthcare Regulations:

- **KVKK (Personal Data Protection Law):** All patient data must be processed in compliance with Turkish data protection regulations, including explicit consent requirements and data minimization principles
- **Ministry of Health Guidelines:** System must adhere to official healthcare protocols and diagnostic standards established by Turkish health authorities
- **Medical Practice Regulations:** Compliance with Turkish medical licensing requirements and professional conduct standards
- **Healthcare Facility Licensing:** Integration requirements with both public and private healthcare facility regulations

International Compliance Standards:

- **HIPAA Compliance:** Healthcare data privacy and security standards for potential international interoperability
- **GDPR Requirements:** European data protection standards for cross-border data handling
- **HL7 FHIR Standards:** Healthcare data exchange format compliance for interoperability
- **ISO 27001 Security Standards:** Information security management system requirements

Government Integration Requirements:

- **e-Devlet Integration Standards:** Mandatory compliance with Turkish e-Government authentication protocols

- **e-Nabız Platform Specifications:** Required adherence to national health platform data formats and security requirements
- **Government Audit Standards:** Compliance with official audit trail and logging requirements for government systems

2.4.2 Hardware Limitations

Server Infrastructure Constraints:

- **Processing Power:** AI diagnostic algorithms require minimum 16-core processors with high computational capacity
- **Memory Requirements:** Minimum 64GB RAM for concurrent AI processing and database operations
- **Storage Capacity:** Minimum 10TB SSD storage for patient data with 10-year retention requirement
- **Network Bandwidth:** Minimum 1Gbps dedicated bandwidth for real-time data processing and external integrations

Client-Side Limitations:

- **Device Compatibility:** System must function on devices with minimum 2GB RAM and 1024x768 screen resolution
- **Browser Constraints:** Limited to modern browsers supporting HTML5, CSS3, and ES6+ JavaScript standards
- **Network Dependencies:** Requires stable internet connection with minimum 1Mbps bandwidth
- **Mobile Constraints:** Responsive design limitations for complex diagnostic interfaces on small screens

Geographic Infrastructure:

- **Data Center Location:** All servers must be physically located within Turkish borders for data sovereignty compliance
- **Network Latency:** Maximum 200ms response time for users across Turkey's geographic regions
- **Redundancy Requirements:** Mandatory backup infrastructure in geographically separate locations within Turkey

2.4.3 Interfaces to Other Applications

Mandatory Government Integrations:

- **e-Devlet Platform:** Required for all user authentication and identity verification processes

- **e-Nabız System:** Mandatory integration for health data synchronization and regulatory reporting
- **Ministry of Health Systems:** Required connection for inspection board reporting and compliance monitoring

Healthcare System Integrations:

- **Hospital Information Systems (HIS):** Integration constraints with existing hospital management software
- **Laboratory Information Systems (LIS):** Required compatibility with various laboratory data formats and protocols
- **Medical Imaging Systems (PACS/DICOM):** Future integration requirements with medical imaging standards

2.4.4 Parallel Operation

Concurrent User Management:

- **Maximum Concurrent Users:** System must support minimum 10,000 simultaneous active users across all roles
- **Resource Allocation:** Automatic load balancing to prevent system degradation during peak usage
- **Session Management:** Isolation of user sessions to prevent data contamination between concurrent operations

AI Processing Constraints:

- **Diagnostic Queue Management:** Maximum 100 concurrent AI diagnostic requests with priority queuing
- **Model Loading:** Constraints on simultaneous AI model access to prevent memory conflicts
- **Training Operations:** AI model retraining must occur during scheduled maintenance windows to avoid interference

Database Concurrency:

- **Lock Management:** Optimistic locking to prevent data conflicts during simultaneous updates
- **Backup Operations:** Scheduled database backups during low-usage periods to minimize performance impact

2.4.5 Audit Functions

Comprehensive Logging Requirements:

- **User Activity Tracking:** Complete audit trail of all user actions including login, data access, and system modifications

- **Data Access Logging:** Detailed logs of all patient data access with timestamp, user identity, and purpose
- **System Change Tracking:** Version control and change logs for all system configurations and updates
- **AI Decision Logging:** Complete record of AI diagnostic processes, inputs, outputs, and confidence levels

Regulatory Audit Compliance:

- **Government Audit Standards:** Compliance with Turkish healthcare audit requirements and data retention policies
- **Inspection Board Access:** Dedicated audit interfaces for Ministry of Health inspection board investigations
- **Compliance Reporting:** Automated generation of regulatory compliance reports and audit summaries
- **Data Integrity Verification:** Cryptographic hash verification for audit log integrity and tamper detection

Performance and Security Audits:

- **Security Event Logging:** Comprehensive logging of all security events, failed login attempts, and suspicious activities
- **Performance Monitoring:** Continuous tracking of system performance metrics and resource utilization
- **Error Tracking:** Detailed logging of system errors, exceptions, and recovery operations
- **Backup Verification:** Regular audit of backup integrity and disaster recovery capabilities

2.4.6 Control Functions

2.4.6.1 Input Validity Control

Data Validation Requirements:

- **Format Validation:** All user inputs must be validated against predefined formats (email addresses, phone numbers, medical IDs)
- **Range Validation:** Numeric inputs must be within acceptable medical ranges (age 0-150, vital signs within normal parameters)
- **Mandatory Field Validation:** Required fields must be completed before form submission with clear error messaging
- **File Upload Validation:** Medical documents and test results must be in approved PDF format with size limitation

Medical Data Validation:

- **Medication Validation:** Drug names and dosages must be verified against approved pharmaceutical databases
- **Diagnostic Code Validation:** All diagnostic codes must comply with ICD-10 or local Turkish medical coding standards
- **Test Result Validation:** Laboratory results must be within acceptable ranges with automatic flagging of critical values

Security Input Controls:

- **SQL Injection Prevention:** All database queries must use parameterized statements to prevent injection attacks
- **Cross-Site Scripting (XSS) Prevention:** All user inputs must be sanitized to prevent malicious script execution
- **File Upload Security:** Uploaded files must be scanned for malware and restricted to approved file types
- **Input Length Limitations:** Maximum character limits enforced to prevent buffer overflow attacks

2.4.6.1.1 Integrity Control

Data Integrity Assurance:

- **Database Constraints:** Foreign key relationships and referential integrity enforcement across all medical data relationships
- **Transaction Atomicity:** All multi-step operations must complete fully or rollback completely to maintain data consistency
- **Checksum Verification:** Critical medical data must include cryptographic checksums for tamper detection
- **Version Control:** Complete versioning of patient records with change tracking and rollback capabilities

System Integrity Monitoring:

- **Configuration Management:** Automated detection of unauthorized system configuration changes
- **Code Integrity:** Digital signatures for all application code with automatic verification during system startup
- **Database Integrity:** Regular consistency checks and automatic repair of database corruption
- **Backup Integrity:** Verification of backup completeness and recoverability through automated testing

2.4.7 Higher-order Language Requirements

Development Language Constraints:

- **Backend Development:** Must use enterprise-grade languages (Java, Python, C#, or Node.js) with strong typing and memory management
- **Frontend Development:** HTML5, CSS3, and modern JavaScript (ES6+) with TypeScript for large-scale development
- **Database Queries:** SQL compliance with support for complex medical data relationships and reporting queries
- **AI/ML Development:** Python with TensorFlow/PyTorch frameworks for machine learning model development and deployment

Code Quality Standards:

- **Documentation Requirements:** Comprehensive code documentation with medical domain-specific comments
- **Testing Coverage:** Minimum 90% code coverage with automated unit, integration, and end-to-end testing
- **Security Standards:** Secure coding practices with regular security code reviews and static analysis
- **Performance Optimization:** Code must meet performance benchmarks for real-time diagnostic processing

2.4.8 Signal Handshake Protocols

External System Communication:

- **e-Devlet Integration:** OAuth 2.0 with mutual TLS authentication for secure government system communication
- **e-Nabız Protocol:** Government-specified secure API protocols with digital certificate authentication
- **Healthcare Standards:** HL7 FHIR RESTful APIs for healthcare data exchange with standard error handling
- **Database Connections:** Encrypted database connections with automatic retry and failover mechanisms

Internal System Communication:

- **Microservices Architecture:** RESTful APIs with JSON payload format and standardized error response codes
- **Real-time Communication:** WebSocket connections for live updates with heartbeat monitoring
- **AI Model Communication:** High-performance gRPC protocols for AI inference requests

and responses

- **Load Balancer Communication:** Health check endpoints with automatic service discovery and routing

2.4.9 Reliability Requirements

System Availability:

- **Uptime Target:** 99.9% availability (maximum 8.76 hours downtime per year)
- **Recovery Time Objective (RTO):** Maximum 4 hours for complete system recovery from major failures
- **Recovery Point Objective (RPO):** Maximum 1 hour of data loss acceptable during disaster scenarios
- **Failover Capability:** Automatic failover to backup systems within 5 minutes of primary system failure

Error Handling and Recovery:

- **Graceful Degradation:** System must continue core functions even when non-critical components fail
- **Automatic Recovery:** Self-healing capabilities for transient errors and network interruptions
- **Data Consistency:** Guaranteed data consistency across all system components during failure scenarios
- **User Notification:** Clear communication to users during system issues with estimated resolution times

2.4.10 Criticality of the Application

Mission-Critical Classification:

- **Life Safety Impact:** System failures could potentially impact patient safety through delayed or incorrect diagnoses
- **Healthcare Continuity:** System unavailability directly affects healthcare service delivery across multiple facilities
- **Regulatory Compliance:** System failures could result in regulatory violations and legal consequences
- **Data Sensitivity:** Protection of highly sensitive personal health information with potential for significant harm if compromised

Critical Function Priorities:

1. **Emergency Access:** Core diagnostic functions must remain available for emergency medical situations
2. **Data Integrity:** Patient health data must be protected and available at all times

3. **Authentication Services:** Secure user access must be maintained to prevent unauthorized data access
4. **Audit Functions:** Complete audit trails must be maintained for regulatory compliance and legal requirements

2.4.11 Safety and Security Considerations

Cybersecurity Requirements:

- **Data Encryption:** AES-256 encryption for all patient data at rest and in transit
- **Access Control:** Multi-factor authentication and role-based access control with regular access reviews
- **Network Security:** Firewalls, intrusion detection systems, and regular security vulnerability assessments
- **Incident Response:** Comprehensive cybersecurity incident response plan with stakeholder notification procedures

Privacy Protection:

- **Data Minimization:** Collection and processing of only necessary patient data for diagnostic purposes
- **Consent Management:** Explicit patient consent tracking for all data usage with granular permission controls
- **Anonymization:** Patient data anonymization for research and statistical analysis purposes
- **Right to Deletion:** Patient rights to request data deletion in compliance with privacy regulations

Physical Security:

- **Data Center Security:** Physical access controls and environmental monitoring for server infrastructure
- **Device Security:** Secure configuration requirements for all client devices accessing the system
- **Backup Security:** Secure storage and transportation of backup media with encryption and access controls
- **Disposal Security:** Secure destruction of hardware containing patient data according to regulatory standards

2.5. Assumptions and Dependencies

The successful implementation and operation of the docWise system relies on several critical assumptions and dependencies that must be maintained throughout the system lifecycle.

2.5.1 External System Dependencies

e-Government (e-Devlet) Platform Dependencies:

- The e-Devlet authentication system must be continuously available for user login and identity verification
- e-Devlet API services must maintain backward compatibility with existing authentication protocols
- Government digital certificates must remain valid and renewable for secure system communication
- e-Devlet user database must be accessible for real-time identity verification during registration and login processes

e-Nabız National Health Platform Dependencies:

- The e-Nabız system must be operational for automatic health data import and synchronization
- e-Nabız API endpoints must remain stable and provide consistent data formats for patient health records
- National health data standards must be maintained for seamless integration with patient medical histories
- e-Nabız platform must support bi-directional data sharing for test result uploads and compliance reporting

Ministry of Health Infrastructure Dependencies:

- MOH inspection board systems must be accessible for case submission and regulatory reporting
- Healthcare regulations and compliance standards must remain consistent during system operation
- Official audit trail requirements must be stable to ensure continuous compliance
- Government network infrastructure must support secure communication protocols

2.5.2 Technology Infrastructure Assumptions

Internet Connectivity Requirements:

- Users must have stable internet connection with minimum 1 Mbps bandwidth for basic functionality
- Healthcare facilities must maintain reliable high-speed internet (minimum 10 Mbps) for

real-time diagnostic operations

- Network connectivity must be available 24/7 for emergency medical situations and critical patient care
- Internet service providers must maintain adequate uptime (99%+) for continuous system access

Client Device Assumptions:

- Users have access to modern computing devices (computers, tablets, or smartphones) capable of running current web browsers
- Patient devices must support standard web technologies including HTML5, CSS3, and JavaScript ES6+
- Healthcare professional workstations must meet minimum performance requirements for complex diagnostic interfaces
- All client devices must be capable of secure HTTPS communication and digital certificate validation

Third-Party Service Dependencies:

- Cloud hosting services must provide 99.9% uptime guarantees with geographically distributed data centers within Turkey
- Database services must maintain ACID compliance and support real-time backup and recovery operations
- Content delivery networks (CDN) must ensure fast page loading across all regions of Turkey
- SSL certificate providers must maintain valid certificates for secure communication protocols

2.5.3 Healthcare System Assumptions

Medical Practice Standards:

- Turkish healthcare protocols and diagnostic procedures must remain consistent with current international medical standards
- Medical professionals must have adequate training in AI-assisted diagnostic tools and digital health platforms
- Healthcare facilities must maintain standard operating procedures compatible with digital workflow integration
- Medical licensing and certification systems must support digital verification through e-Government platforms

Data Quality Assumptions:

- e-Nabız health data must be accurate, complete, and regularly updated by healthcare providers

- Patient medical histories must be comprehensive enough to support AI diagnostic analysis
- Laboratory test results must be standardized and compatible with digital import processes
- Medical imaging data must conform to standard formats (DICOM) for future integration capabilities

User Behavior Assumptions:

- Patients will actively participate in maintaining their health data and following digital health recommendations
- Medical doctors will appropriately balance AI diagnostic suggestions with clinical judgment and experience
- Hospital administrators will properly configure and maintain system integration with existing healthcare workflows
- All users will comply with data privacy requirements and security protocols

2.5.4 Regulatory and Legal Dependencies

Healthcare Regulation Stability:

- Turkish healthcare laws and regulations (KVKK, MOH guidelines) must remain stable during system implementation
- International healthcare data standards (HL7 FHIR, DICOM) must maintain compatibility with Turkish requirements
- Data privacy regulations must continue to support the current approach to patient consent and data processing
- Medical malpractice and liability frameworks must accommodate AI-assisted diagnostic tools

Government Policy Dependencies:

- Digital transformation initiatives in Turkish healthcare must continue to support integrated health information systems
- Government funding and support for national health platform improvements must be sustained
- Cross-ministry coordination between health, technology, and digital governance agencies must be maintained
- National cybersecurity policies must align with healthcare data protection requirements

2.5.5 Operational Assumptions

Staffing and Training:

- Healthcare facilities must provide adequate staff training for digital health platform adoption
- Technical support personnel must be available for system maintenance and user assistance
- Medical professionals must allocate time for learning and adapting to AI-assisted diagnostic workflows
- Administrative staff must be trained in new digital processes for patient management and compliance reporting

Business Continuity:

- Healthcare facilities must maintain backup procedures for system outages or technical failures
- Alternative communication methods must be available when digital systems are unavailable
- Emergency medical protocols must account for potential system unavailability during critical situations
- Data backup and recovery procedures must be regularly tested and validated

Performance and Scalability:

- System usage will grow gradually, allowing for incremental capacity planning and infrastructure scaling
- AI diagnostic accuracy will improve over time through continuous learning and model updates
- User adoption rates will be sufficient to justify ongoing system development and maintenance costs
- Healthcare outcome improvements will be measurable and demonstrate system value

2.5.6 Financial and Resource Dependencies

Funding Assumptions:

- Adequate financial resources must be available for ongoing system maintenance, updates, and improvements
- Healthcare facilities must budget for necessary infrastructure upgrades and staff training
- Government funding must support national health platform integration and regulatory compliance activities
- Insurance coverage and billing systems must accommodate digital health platform integration

Vendor and Service Provider Dependencies:

- Technology vendors must provide ongoing support and maintenance for critical system components
- Cloud service providers must maintain service level agreements and support Turkey-specific data sovereignty requirements
- Software license providers must continue to offer necessary development and deployment tools
- Security service providers must maintain current threat protection and monitoring capabilities

2.6. Apportioning of Requirements

This section defines how requirements will be distributed across different system releases and implementation phases, ensuring systematic delivery of functionality while maintaining system integrity and user satisfaction.

2.6.1 Release Planning Strategy

Phase 1: Core System Implementation (v1.0) *Target Timeline: 6 months Priority: Critical Foundation*

Essential Authentication and User Management:

- e-Devlet integration for secure user authentication (Patients, Doctors, Administrators)
- Basic user profile management and preferences
- Role-based access control and security framework
- Session management and audit logging

Basic AI Diagnostic Functionality:

- Core AI diagnostic engine with initial medical knowledge base
- Simple symptom input and basic diagnostic suggestion generation
- Doctor review and approval/rejection workflow with justification requirements
- Initial diagnostic accuracy tracking and reporting

Fundamental Data Management:

- e-Nabız integration for basic health data import
- Patient medical history viewing and basic data display
- Simple treatment plan creation and storage
- Basic PDF report generation for diagnoses and treatment plans

Minimum Viable Administrative Functions:

- Basic patient registration and appointment management

- Simple laboratory test request and result upload
- Elementary compliance tracking and audit trail generation
- Basic system monitoring and user activity logging

2.6.2 Subsequent Release Priorities

Phase 2: Enhanced Functionality (v1.5) *Target Timeline: 3 months after v1.0 Priority: User Experience and Integration*

Advanced Patient Features:

- Comprehensive graphical health analysis and trend visualization
- AI-generated lifestyle recommendations based on health data
- Enhanced health history management with filtering and search capabilities
- Mobile-responsive interface optimization for patient portal

Improved Clinical Workflow:

- Advanced diagnostic parameter selection and analysis
- Enhanced treatment plan templates and standardization
- Improved laboratory test integration with automated result processing
- Advanced symptom documentation with medical terminology support

Hospital Administration Enhancement:

- Insurance policy integration and automated pricing calculation
- Advanced appointment scheduling and patient flow management
- Comprehensive billing and payment processing for private healthcare
- Enhanced reporting and analytics for hospital performance monitoring

Phase 3: Advanced Analytics and Intelligence (v2.0) *Target Timeline: 6 months after v1.5 Priority: AI Enhancement and Analytics*

Advanced AI Capabilities:

- Machine learning model improvement with continuous learning algorithms
- Multi-parameter diagnostic analysis with confidence scoring
- Predictive health analytics and early warning systems
- Advanced diagnostic accuracy tracking with performance analytics

Comprehensive Administrative Oversight:

- Advanced doctor performance monitoring and statistical analysis
- Automated necessity analysis for treatment plans with threshold management
- Intelligent investigation management and malicious practice detection

- Comprehensive regulatory compliance reporting and audit trail management

Enhanced Integration and Interoperability:

- Advanced e-Nabız integration with real-time data synchronization
- Ministry of Health inspection board full integration with case management
- Third-party healthcare system integration capabilities
- Advanced security and encryption for all data communications

2.6.3 Future Enhancement Roadmap

Phase 4: Advanced Features (v2.5) *Target Timeline: 12 months after v2.0 Priority: Advanced Capabilities*

Telemedicine Integration:

- Remote consultation capabilities with video conferencing
- Remote diagnostic support for rural and underserved areas
- Mobile health monitoring integration with wearable devices
- Advanced patient self-service capabilities

Advanced Analytics and Reporting:

- Population health analytics and epidemiological tracking
- Advanced research capabilities with anonymized data analysis
- Predictive modeling for disease outbreak and public health monitoring
- Advanced business intelligence and decision support analytics

Enhanced Collaboration Features:

- Doctor-to-doctor consultation and second opinion capabilities
- Multi-disciplinary team collaboration tools
- Medical education and training integration
- Advanced case study and knowledge sharing platforms

2.6.4 Requirement Distribution Criteria

Critical Path Requirements (Must Have - Phase 1):

- User authentication and basic security (100% in v1.0)
- Core AI diagnostic functionality (80% in v1.0, 20% in v1.5)
- Essential data management and e-Nabız integration (90% in v1.0, 10% in v1.5)
- Basic administrative and compliance functions (70% in v1.0, 30% in v1.5)

Important Requirements (Should Have - Phase 2):

- Enhanced user experience and interface improvements (100% in v1.5)

- Advanced clinical workflow management (80% in v1.5, 20% in v2.0)
- Comprehensive hospital administration features (90% in v1.5, 10% in v2.0)
- Enhanced integration capabilities (70% in v1.5, 30% in v2.0)

Desirable Requirements (Could Have - Phase 3+):

- Advanced AI and machine learning capabilities (60% in v2.0, 40% in v2.5)
- Comprehensive analytics and reporting (50% in v2.0, 50% in v2.5)
- Advanced collaboration and telemedicine features (20% in v2.0, 80% in v2.5)
- Research and population health capabilities (10% in v2.0, 90% in v2.5)

2.6.5 Dependencies and Risk Management

Cross-Phase Dependencies:

- AI model training requires data from Phase 1 implementation before Phase 2 enhancement
- Advanced analytics capabilities depend on sufficient user adoption and data accumulation
- Regulatory compliance features must be implemented before advanced administrative functions
- Security and audit frameworks must be established in Phase 1 before subsequent feature additions

Risk Mitigation Strategies:

- **Technical Risk:** Parallel development tracks for critical components with fallback options
- **Integration Risk:** Staged integration testing with external systems throughout all phases
- **User Adoption Risk:** Comprehensive training programs and change management support
- **Regulatory Risk:** Continuous compliance monitoring and legal review throughout development

Success Criteria for Each Phase:

- **Phase 1:** 75% AI diagnostic accuracy, 99% system uptime, successful e-Devlet/e-Nabız integration
- **Phase 2:** 80% user satisfaction rating, 50% reduction in administrative overhead, enhanced diagnostic workflow efficiency
- **Phase 3:** 85% AI diagnostic accuracy, comprehensive regulatory compliance, measurable healthcare outcome improvements
- **Phase 4:** Advanced analytics capabilities, telemedicine readiness, research platform

functionality

3. Specific Requirements

3.1. Functions

Use Case ID	UC_01 – Patient Sign up
Description	The patient clicks the “Patient Sign Up” option and registers to the system via e-Government.
Actors	Patient
Preconditions	The patient must be able to access the e-Government system.
Post Conditions	After the registration process is completed, the patient is defined as a user and an account is created.
Main Success Scenario	1. The patient accesses the website. 2. Clicks the “Patient Sign Up” button. 3. The system redirects the patient to the e-Government verification screen. 4. The patient logs in with e-Government credentials. 5. The system successfully completes identity verification and creates the patient account. 6. A success message is displayed.
Alternative Flow	4a. If the patient cancels the e-Government identity verification, the system ends the registration process and redirects to the homepage.
Exceptional Flow	3a. If the e-Government system is unavailable or a connection error occurs, the system shows an error message to the user and cancels the registration process.
Rules	**R1.** Each user can only register once. **R2.** The registration process cannot be completed without e-Government identity verification.

Use Case ID	UC_02 – Patient Login
Description	The patient clicks the “Patient Login” option and logs into the system via e-

Use Case ID	UC_02 – Patient Login
	Government.
Actors	Patient
Preconditions	The patient must already be registered in the system and must be able to access the e-Government system.
Post Conditions	The patient successfully logs into the system and is redirected to the user interface.
Main Success Scenario	1. The patient goes to the web interface. 2. Clicks the “Patient Login” option. 3. The system redirects the user to the e-Government screen. 4. The patient logs in using e-Government credentials. 5. If identity verification is successful, the patient is redirected to the user panel.
Alternative Flow	4a. If the patient closes the identity verification screen, the system returns to the homepage.
Exceptional Flow	3a. If the e-Government system is down or a connection error occurs, the system shows an error message and login is not possible.
Rules	**R1.** Only previously registered users can log in. **R2.** Login must be performed through e-Government identity verification.

Use Case ID	UC_03 – Doctor Sign Up
Description	The doctor clicks the “Doctor Sign Up” option and registers to the system via e-Government.
Actors	Doctor
Preconditions	The doctor must have authorization for **e-Government doctor login**.
Post Conditions	The doctor successfully registers to the system and a user account is created.
Main Success Scenario	1. The doctor accesses the web interface. 2. Clicks the “Doctor Sign Up” option. 3. The system redirects the doctor to the e-Government verification page. 4. The doctor logs in with e-Government credentials. 5. If identity verification is successful, the system registers the doctor and displays a success message.
Alternative Flow	4a. If the doctor closes the e-Government verification screen, the system cancels the registration and redirects to the login page.
Exceptional Flow	3a. If the e-Government system is not reachable, the user is informed that the system is temporarily unavailable.
Rules	**R1.** Duplicate registration with the same doctor credentials is not allowed. **R2.** Identity verification must be done through e-Government integration.

Use Case ID	**UC_04 – Doctor Login**
Description	The doctor clicks the “Doctor Login” option and logs into the system via e-Government.
Actors	Doctor
Preconditions	The doctor must already be registered in the system and be able to access the e-Government doctor login system.
Post Conditions	The doctor successfully logs into the system and is redirected to the doctor panel.
Main Success Scenario	1. The doctor accesses the web interface. 2. Clicks the “Doctor Login” option. 3. The system redirects the user to the e-Government verification screen. 4. The doctor logs in using e-Government credentials. 5. If identity verification is successful, the doctor is redirected to the user panel.
Alternative Flow	4a. If the doctor closes the identity verification screen, the system redirects to the homepage.
Exceptional Flow	3a. If the e-Government system is not working or there is a connection problem, the system displays an error message and cancels the login process.
Rules	**R1.** Only doctors who are already registered can log in. **R2.** Identity verification must be performed only via the e-Government infrastructure.

Use Case ID	**UC_06 – Edit e-mail or phone number**
Description	The user can edit the phone number and e-mail address registered in the system.
Actors	Patient, Doctor, Admin
Preconditions	The user must be logged into the system.
Post Conditions	The user's new phone number and/or e-mail address is successfully updated in the system.
Main Success Scenario	1. User logs in. 2. Goes to the profile settings page. 3. Edits the phone number and/or e-mail fields. 4. Clicks the “Save” button. 5. The system validates and updates the information. 6. A message is displayed: “Your information has been successfully updated.”
Alternative Flow	4a. The user edits only one field and leaves the other empty; the system only updates the edited field.
Exceptional Flow	5a. If the entered e-mail or phone number format is invalid, the system displays an error message: “Invalid e-mail address”, “Invalid phone number”.

Use Case ID	UC_06 – Edit e-mail or phone number
Rules	**R1.** E-mail address must be in valid format. **R2.** Phone number must be 10 digits (excluding leading “0”).

Use Case ID	UC_07 – Choose Theme
Description	The user can change the interface theme (e.g., dark mode).
Actors	Patient, Doctor, Admin
Preconditions	The user must be logged into the system.
Post Conditions	The selected theme is saved based on user preference and the interface theme is updated.
Main Success Scenario	1. User logs in. 2. Navigates to the “Theme Selection” section from the settings menu. 3. Selects one of the available themes (e.g., “Dark Theme”). 4. Clicks the “Save” button. 5. The system saves the new theme preference and instantly updates the interface.
Alternative Flow	3a. If the user exits without saving after changing the theme, the previous theme preference is retained.
Exceptional Flow	5a. If an error occurs during the update process, the system displays a warning: “Theme preference could not be saved.”
Rules	**R1.** Theme preference is stored separately for each user. **R2.** The selected theme is automatically applied when the user logs into the system.

Use Case ID	UC_08 – Choose Language
Description	The user can select a preferred interface language (e.g., Turkish or English).
Actors	Patient, Doctor, Admin
Preconditions	The user must be logged into the system.
Post Conditions	The selected language is saved and the interface language is updated.
Main Success Scenario	1. User logs into the system. 2. Navigates to the “Language Preference” tab from the settings menu. 3. Selects one of the options: “Turkish” or “English”. 4. Clicks the “Save” button. 5. The system automatically updates the interface according to the

Use Case ID	UC_08 – Choose Language
	selected language.
Alternative Flow	3a. If the user changes the language preference but exits without saving, the previous language preference is retained.
Exceptional Flow	5a. If an error occurs during the update, the system displays a warning: “Language preference could not be saved.”
Rules	**R1.** The user's selected language preference should be automatically applied during the next login. **R2.** Supported languages must be defined in the system (e.g., only Turkish and English).

Use Case ID	UC_09 – Integrate e-Nabız data
Description	The patient connects their e-Nabız account to automatically transfer health data into the system.
Actors	Patient
Preconditions	The patient must be logged into the system and have an active account in the e-Nabız system.
Post Conditions	The patient’s e-Nabız data is successfully transferred into the system and becomes analyzable by the AI module.
Main Success Scenario	1. Patient logs into the system. 2. Clicks on the “e-Nabız Integration” tab. 3. The system performs the e-Nabız integration using the patient's e-Government credentials. 4. The system retrieves e-Nabız data and displays the message: “Data successfully transferred.”
Alternative Flow	4a. If the patient closes the e-Nabız connection screen, the data transfer is canceled and the user is informed.
Exceptional Flow	5a. If the e-Nabız system is unreachable, the user sees the message “The e-Nabız system is currently unavailable” and the process is terminated.
Rules	**R1.** Patient data can only be transferred into the system with the user's explicit consent. **R2.** Transferred data is private and accessible only to the associated user.

Use Case ID	UC_10 – View Diagnosis & Treatment Plans
Description	The patient views previous diagnosis and treatment plans.
Actors	Patient

Preconditions	The patient must be logged into the system and have recorded past diagnoses/history in the system.

Use Case ID	UC_10 – View Diagnosis & Treatment Plans
Post Conditions	The patient successfully views past diagnoses and treatment plans; they can be used for further analysis.
Main Success Scenario	1. Patient logs into the system. 2. Clicks the “My Treatments” or “My Health History” tab. 3. The system lists the patient's previously received diagnoses and treatment data. 4. The patient clicks on the desired record. 5. The details page shows the corresponding diagnosis and treatment plan.
Alternative Flow	3a. On the list screen, the patient applies a date or category filter.
Exceptional Flow	3b. If no diagnosis/treatment record exists in the patient's history, the system displays the message: “No record found.”
Rules	**R1.** The user can only view their own diagnosis/treatment history. **R2.** All data is in read-only mode and cannot be modified.

Use Case ID	UC_11 – Download as PDF
Description	The patient can download past diagnosis and treatment plans as a PDF file.
Actors	Patient
Preconditions	The patient must be logged into the system and have at least one diagnosis/medical record in the system.
Post Conditions	The selected diagnosis and treatment plan is downloaded to the patient’s device as a PDF file.
Main Success Scenario	1. Patient logs into the system. 2. Navigates to the “My Past Diagnoses” page. 3. Selects the record they want to view. 4. Clicks the “Download as PDF” button. 5. The system generates the PDF and downloads it to the user’s device.
Alternative Flow	*(No alternative flow specified)*
Exceptional Flow	5a. If an error occurs while generating the PDF file, the system displays a warning: “PDF file could not be generated.”
Rules	**R1.** The PDF output can only be downloaded by the relevant user. **R2.** The PDF file must include the date, diagnosis, and treatment plan details.

Use Case ID	**UC_12 – Retrieve Graphical Analyses** ***(This section generates graphical analysis based on the patient's past blood values, as referenced in the mockup.)***
Description	The patient receives graphical analyses based on past health data.
Actors	Patient
Preconditions	The patient must be logged into the system and must have past health data stored in the system.
Post Conditions	The patient accesses the graphical analysis screen and can view their health data in a visualized format.
Main Success Scenario	1. Patient logs into the system. 2. Clicks on the "Health Analysis" tab. 3. The system processes the patient's past lab results, diagnoses, and treatment data. 4. Visualized graphs (e.g., time series, bar charts) are generated. 5. The patient reviews health trends and past conditions via the graphs.
Alternative Flow	3a. The user can select the type of analysis (e.g., only blood tests, only diagnosis history, etc.), and the graphs are updated accordingly.
Exceptional Flow	3b. If the user does not have enough health data in the system for analysis, the system displays the warning: "Insufficient health data to generate analysis."
Rules	**R1.** Graphs are for visual purposes only and do not constitute a medical diagnosis.

Use Case ID	**UC_13 – Receive AI Life Support Suggestion**
Description	The patient receives an AI-based life support suggestion based on graphical analyses.
Actors	Patient
Preconditions	The patient must be logged into the system and must have past health data suitable for analysis in the system.
Post Conditions	A life support suggestion generated by the AI model is presented to the user.
Main Success Scenario	1. Patient logs into the system. 2. Navigates to the "Health Analysis" or "Suggestions" tab. 3. The system sends the analysis results based on past data to the AI module. 4. The AI model evaluates the user's data and generates personalized life support suggestions.
Alternative Flow	*(None specified)*
Exceptional Flow	4b. If the AI model cannot generate a suggestion due to insufficient data, the system displays the message: "Insufficient data, suggestion could not be generated."

Rules	**R1.** Suggestions are for informational purposes only and do not replace medical diagnosis or treatment. **R2.** Data is not sent to the AI model without the user's consent.

Use Case ID	**UC_14 – Upload Health Insurance Policy**
Description	The patient uploads their current insurance policy to the system.
Actors	Patient
Preconditions	The patient must be logged into the system. The file to be uploaded must be in .pdf format.
Post Conditions	• All uploaded policies are successfully stored in the system. • Each policy becomes accessible for verification by the private hospital administration.
Main Success Scenario	1. Patient logs into the system. 2. Clicks the "Insurance Policy" tab. 3. Clicks the "Upload New Policy" button. 4. Selects one or more policy files in .pdf format from their device. 5. Clicks the "Upload" button. 6. The system saves each policy successfully and displays a confirmation message.
Alternative Flow	4a. If the user clicks the upload button without selecting a file, the system displays a warning: "Please select a file."
Exceptional Flow	6a. If the network disconnects during upload, the system displays: "Upload failed, please try again."
Rules	**R1.** Only files in .pdf format are accepted. **R2.** Each user can upload more than one active insurance policy. **R3.** The same type of insurance policy cannot be uploaded more than o

Use Case ID	**UC_15 – Register Patient (Government)**
Description	The hospital administration registers a patient who visits a public hospital for an appointment with a relevant doctor.
Actors	Hospital Administration
Preconditions	The patient must have applied to a public hospital and the relevant doctor must be registered in the system.
Post Conditions	The patient becomes visible in the system with an active appointment registration for the relevant doctor.

Main Success Scenario	1. The patient physically applies to the public hospital. 2. The hospital administration logs into the system to register the patient. 3. The relevant doctor is selected. 4. The system retrieves the patient’s identity information and creates a new appointment record. 5. The appointment becomes visible on the doctor’s system screen.

Use Case ID	UC_15 – Register Patient (Government)
Alternative Flow	*(None specified)*
Exceptional Flow	4a. If the doctor is not active in the system, the system displays the warning: “The relevant doctor is currently unavailable,” and the process is canceled.
Rules	**R1.** Multiple appointment records cannot be created for the same patient with the same doctor on the same day. **R2.** Appointment records can only be created by authorized hospital administrators.

Use Case ID	UC_16 – Price the Examination
Description	The hospital administration prices the examination for a patient applying to a private hospital. If there is a valid insurance policy uploaded to the system, pricing is based on that policy; otherwise, the standard fee schedule is applied.
Actors	Hospital Administration
Preconditions	(If applicable) The insurance policy must have been previously uploaded to the system.
Post Conditions	• The **system** calculates the price based on the patient's insurance policy if available, or applies the standard fee schedule. • The **patient** is informed of the amount they need to pay.
Main Success Scenario	1. Patient applies to the private hospital. 2. **Hospital administration** checks the insurance policy uploaded to the system. 3. a. If a valid policy exists, the **system** identifies the procedures covered by the policy. b. If no policy exists, the **system** applies the standard fee schedule for the patient. 4. **System** deducts the covered amount and calculates the total examination fee. 5. Patient sees the **net amount** to be paid in the system.
Alternative Flow	3a. Hospital administration manually reviews the uploaded policy and revises the system’s pricing if necessary. 3b. If no policy is found, the user is shown a message: “No insurance found, standard pricing will be applied.”
Exceptional Flow	4a. If the insurance policy is invalid or unreadable in the system, a message is displayed: “Insurance information not found,” and the operation is canceled.
Rules	**R1.** Pricing is based on the insurance policy if valid, otherwise, the standard private hospital tariff is applied. **R2.** The system can automatically calculate discounts only for recognized insurance companies and their covered services. **R3.** Patients without insurance can receive service at standard prices without any restriction.
Use Case ID	UC_17 – Register Patient (Special)
Description	The hospital administration registers the examination of a patient who

Use Case ID	UC_16 – Price the Examination
	accepts the pricing for a private hospital.
Actors	Hospital Administration
Preconditions	The patient must have accepted the pricing proposed by the private hospital.
Post Conditions	The patient becomes visible in the system with an active appointment registration for the relevant doctor.
Main Success Scenario	1. The patient applies to the private hospital and the system performs pricing. 2. The patient approves the specified price. 3. Hospital administration logs into the system. 4. Based on the approved pricing, the system presents the option to create a new appointment. 5. Hospital administration confirms the approval, and the system creates the appointment. 6. The appointment becomes visible on the doctor's system screen.
Alternative Flow	*(None specified)*
Exceptional Flow	5a. If the approved price information has been deleted or updated in the system, it displays: “Invalid price information” and the process is canceled.
Rules	**R1.** No appointment can be created without explicit price approval by the patient. **R2.** Appointments can only be created by the hospital administration.

Use Case ID	UC_18 – Register Patient for Test (Government)
Description	The hospital administration registers a laboratory test for a patient in a public hospital.
Actors	Hospital Administration
Preconditions	The patient must have applied to a public hospital and a test request must have been made by a doctor.
Post Conditions	An active laboratory test record is created for the patient in the system.
Main Success Scenario	1. The doctor submits a test request for the patient. 2. Hospital administration logs into the system. 3. From the admin panel, clicks the “Laboratory Registration” tab. 4. Selects the relevant patient. 5. Enters the requested tests and creates the lab registration. 6. The system displays a success message confirming the registration.
Alternative Flow	*(None specified)*
Exceptional Flow	5a. If tests cannot be entered into the system or if a connection error occurs, the system does not save the record and displays the message: “Registration failed.”
Rules	**R1.** A patient cannot be registered for the same test type more than once in the same day. **R2.** Laboratory records can only be initiated by hospital administration. **R3.** Only the tests requested by the doctor can be registered.

Use Case ID	UC_19 – Price the Tests
Description	The hospital administration calculates the pricing of requested tests in a private hospital based on the patient's uploaded insurance policy if available, or applies standard pricing if no insurance exists.
Actors	Hospital Administration
Preconditions	The patient must be registered in the system. A test request must have been made by a doctor. (If applicable) A valid insurance policy may be uploaded in the system.
Post Conditions	The system calculates the insurance coverage rate and the final amount payable by the patient based on the insurance policy (if any) or standard tariff. This information is displayed to the patient.
Main Success Scenario	1. The doctor submits a test request for the patient. 2. Hospital administration accesses the “Test Pricing” tab. 3. Enters the test details. 4. a. If a valid insurance policy exists, the system calculates the covered portion of the test costs based on the policy. b. If no policy exists, the system applies standard test pricing. 5. The system calculates the amount the patient must cover and displays it.
Alternative Flow	3a. The hospital administration uses predefined templates to select frequently used test combinations instead of manual entry.
Exceptional Flow	4a. If the system connection is lost or the data cannot be read while processing, the system displays: “Calculation failed, please try again.”
Rules	**R1.** If a valid insurance policy exists, the payment rate is calculated based on it. **R2.** If no policy exists, standard test pricing is applied. **R3.** Test pricing can only be edited by the hospital administration.

Use Case ID	UC_20 – Register Patient for Test (Special)
Description	The hospital management creates a laboratory record for the patient who accepts the pricing for a private hospital test.
Actors	Hospital Management
Preconditions	The test request must be entered by the doctor, and the patient must have accepted the pricing.
Post Conditions	An active laboratory record containing the relevant tests is created for the patient in the system.
Main Success Scenario	1. The doctor makes a test request. 2. The system performs pricing for the tests based on the insurance policy status. 3. The patient accepts the pricing. 4. Hospital management clicks on the test registration tab. 5. The requested test details are entered and the record is created. 6. The system displays the message “Laboratory record successfully created.”
Alternative Flow	–

Use Case ID	UC_20 – Register Patient for Test (Special)
Exceptional Flow	5a. If the insurance policy is expired or invalid, the system displays a “Policy invalid” warning and the record is not created.
Rules	**R1.** A test record can only be created if price approval has been received. **R2.** The same test cannot be recorded twice for the same day. **R3.** Only the tests requested by the doctor are entered into the system.

Use Case ID	UC_21 – Share test results with the e-Nabız system
Description	The hospital management transfers all uploaded test data to the e-Nabız system for public hospitals.
Actors	Hospital Management
Preconditions	The test must have been performed and the laboratory results must be uploaded to the system. The patient's e-Nabız connection must be active.
Post Conditions	All test data is successfully transferred to the e-Nabız system.
Main Success Scenario	1. The laboratory uploads the test results to the system. 2. The hospital management logs into the system. 3. From the admin panel, the “Transfer e-Nabız Data” tab is opened. 4. The system lists the uploaded test results. 5. The administrator approves the transfer. 6. The system sends the data to the e-Nabız system and displays a “Transfer successful” message.
Alternative Flow	–
Exceptional Flow	6a. If the system cannot access e-Nabız, it shows a “Data transfer failed, please try again” warning.
Rules	**R1.** Previously sent data cannot be sent again.

Use Case ID	UC_22 – View Patient List
Description	The doctor views the patient list.
Actors	Doctor
Preconditions	The doctor must be logged into the system and there must be active patient records.
Post Conditions	The doctor views the patient list to create an examination or treatment plan.

Main Success Scenario	1. The doctor logs into the system. 2. From the main panel, the doctor clicks on the “Patient List” tab. 3. The system retrieves the list of patients assigned to the doctor. 4. The doctor can filter the patients or click on a patient to view their details.

Use Case ID	UC_22 – View Patient List
Alternative Flow	3a. The doctor filters to display only patients within a specific date range.
Exceptional Flow	3b. If no patients are assigned, the system displays the warning “No registered patients found.”
Rules	**R1.** The doctor can only view patients assigned to them. **R2.** The patient list can only be viewed through an active user session.

Use Case ID	UC_23 – Start Treatment Process
Description	The doctor selects the next patient from the list and starts the treatment process.
Actors	Doctor
Preconditions	The doctor must be logged into the system and must have access to the patient list screen. The selected patient must have an active examination record.
Post Conditions	The treatment process is started; the system redirects to the medical procedures page for the patient.
Main Success Scenario	1. The doctor logs into the system. 2. Clicks on the “Patient List” tab. 3. The next patient appears on the list. 4. The doctor selects the patient. 5. The system redirects to the patient’s treatment screen. 6. The doctor begins to enter medical information.
Alternative Flow	3a. The doctor can sort patients by priority (e.g., patients over 65 or pregnant women) and choose a non-next patient early.
Exceptional Flow	4a. If the selected patient’s record has been deleted or marked as passive, the system displays a “Patient record invalid” warning.
Rules	**R1.** Only patients with an active examination record can be selected. **R2.** A patient can be processed on the same day by only one doctor.

Use Case ID	UC_24 – Input Patient Symptoms
Description	The doctor enters the patient’s current symptoms into the system.
Actors	Doctor
Preconditions	The doctor must have selected the patient and started the examination process.

Post Conditions	The patient's symptom information is saved in the system and becomes available for use by the artificial intelligence module.

Use Case ID	UC_24 – Input Patient Symptoms
Main Success Scenario	1. The doctor accesses the examination screen. 2. Clicks on the symptom entry field. 3. Enters the current symptoms reported by the patient into the relevant section. 4. After completing the entry, clicks the "Save" button. 5. The system successfully saves the data.
Alternative Flow	3a. The doctor selects predefined symptoms from the system's symptom list for automatic entry.
Exceptional Flow	5a. If the system connection is lost or verification fails, the system displays a "Symptoms could not be saved" warning.
Rules	**R1.** Symptoms can only be entered for patients with an active examination session. **R2.** Entered symptoms are reflected in the patient record and cannot be deleted; only updated.

Use Case ID	UC_25 – Submit Test Requests to Management
Description	The doctor notifies hospital management to request lab tests for the patient, if necessary.
Actors	Doctor, Hospital Management
Preconditions	The doctor must have started the examination process for the patient and determined the need for tests.
Post Conditions	The hospital management is informed to create a test record for the patient based on the doctor's request.
Main Success Scenario	1. The doctor logs into the system. 2. During the examination, the doctor identifies the need for laboratory tests. 3. The doctor clicks the "Test Request" button and selects the necessary tests to send the request. 4. The system sends the test request to the hospital management panel. 5. The hospital management receives a notification message.
Alternative Flow	3a. The doctor selects a predefined test template in the system to create a quick request.
Exceptional Flow	4a. Due to a system error, if the notification cannot be sent, the doctor receives a "Request could not be sent" warning and the request is not saved.
Rules	**R1.** Test requests can only be submitted for patients currently undergoing an active examination.

Use Case ID	UC_26 – View Test Results
Description	The doctor views the test results uploaded by the hospital laboratory.
Actors	Doctor
Preconditions	The test results must have been uploaded to the system by the laboratory, and the doctor must be assigned to the patient.
Post Conditions	The doctor successfully views the patient's test results in the system.
Main Success Scenario	1. The doctor logs into the system. 2. Selects the relevant patient from the “Patient List” screen. 3. Clicks on the “Test Results” tab. 4. The system lists the test results uploaded by the laboratory. 5. The doctor views the results in detail.
Alternative Flow	3a. The doctor filters the results to view only tests from a specific date range or specific types of tests.
Exceptional Flow	4a. If no test results have been uploaded yet, the system displays the warning “No test results found for this patient.”
Rules	**R1.** The doctor can only access the test results of their own patients. **R2.** The displayed data is read-only and cannot be modified.

Use Case ID	UC_27 – Request for AI-based Diagnosis
Description	The doctor creates a diagnosis request from the AI module by selecting the relevant parameters (such as symptoms, test results, and imaging results) for AI-based diagnosis.
Actors	Doctor
Preconditions	The patient’s symptoms, test results, and, if available, imaging data must have been entered into the system. The AI module must be accessible.
Post Conditions	The AI module generates a diagnosis suggestion based on the given data and presents it to the doctor.
Main Success Scenario	1. The doctor starts the patient examination. 2. Enters the patient’s symptoms, test results, and imaging data into the system or selects existing records. 3. Clicks on the “AI Diagnosis Request” tab. 4. Enters the parameters to be used. 5. Clicks the “Create Diagnosis Request” button. 6. The system sends the data to the AI module. 7. The diagnosis result received from the AI model is displayed to the doctor.
Alternative Flow	6a. If the AI module fails to respond due to a system error, the system displays the warning “Diagnosis could not be generated, please try again.”
Rules	**R1.** The doctor can only request an AI diagnosis using data already saved in the system.

Use Case ID	UC_27 – Request for AI-based Diagnosis
	R2. The AI diagnosis is a suggestion; the final decision belongs to the doctor.

Use Case ID	UC_28 – View AI Diagnosis
Description	The doctor views the AI-generated diagnosis suggestion.
Actors	Doctor
Preconditions	The doctor must have previously created an AI diagnosis request for the patient using relevant data.
Post Conditions	The doctor successfully views the AI-generated diagnosis suggestion.
Main Success Scenario	1. The doctor logs into the system. 2. Clicks on the “AI Diagnosis Results” tab. 3. The system lists previously created diagnosis requests and their results. 4. The doctor selects the relevant patient and views the diagnosis details.
Alternative Flow	3a. The doctor filters the results to view only those from a specific date or with a specific status.
Exceptional Flow	3b. If the diagnosis suggestion has not been generated yet or a system error has occurred, the system displays the warning “Diagnosis result could not be generated.”
Rules	**R1.** The doctor can only view AI diagnosis suggestions for requests they have created. **R2.** The AI diagnosis is for informational purposes only; the doctor can reject it with justification.

Use Case ID	UC_29 – Doctor Accepts the Diagnosis
Description	The doctor approves the diagnosis suggestion provided by the AI.
Actors	Doctor
Preconditions	A diagnosis suggestion generated by the AI must exist, and the doctor must be logged into the system.
Post Conditions	The AI suggestion is approved by the doctor and processed into the patient's record with the doctor's confirmation.
Main Success Scenario	1. The doctor accesses the “AI Diagnosis Results” screen in the system. 2. Selects the relevant patient and the AI diagnosis suggestion. 3. Reviews the diagnosis suggestion. 4. Clicks the “Approve” button. 5. The system records the suggestion as approved in the patient's file and sends a confirmation message.

Use Case ID	UC_29 – Doctor Accepts the Diagnosis
Alternative Flow	3a. The doctor adds a note to the suggestion and approves it. The note is also recorded in the system.
Exceptional Flow	4a. If the record cannot be saved due to a system error, the system displays the warning: “Approval failed, please try again.”
Rules	**R1.** The approved diagnosis is recorded only through the doctor’s user account. **R2.** An approved diagnosis cannot be modified later; however, new diagnoses can be added.

Use Case ID	UC_30 – Submit Rejection Reason
Description	The doctor rejects the AI's diagnosis suggestion with a valid reason.
Actors	Doctor
Preconditions	There must be a diagnosis suggestion generated by the AI, and the doctor must be logged into the system.
Post Conditions	The rejected diagnosis is processed into the patient’s record along with the reason for rejection.
Main Success Scenario	1. The doctor logs into the system. 2. Navigates to the “AI Diagnosis Results” screen. 3. Selects the relevant patient and diagnosis suggestion. 4. Clicks the “Reject” button. 5. The system opens a text field for rejection reasons. 6. The doctor writes the reason and confirms the action. 7. The system records the rejected diagnosis and the reason in the patient’s file.
Alternative Flow	4a. While rejecting the suggestion, the doctor can also define an alternative diagnosis by entering a new note.
Exceptional Flow	6a. If no reason is entered, the rejection process cannot proceed; the system displays: “Please enter a reason.”
Rules	**R1.** Diagnosis suggestions cannot be rejected without a reason. **R2.** The rejected diagnosis is marked as “invalid diagnosis” in the patient’s record.

Use Case ID	UC_31 – Submit Treatment Plan
Description	The doctor prepares the patient's treatment plans and submits them to the system.
Actors	Doctor
Preconditions	The doctor must have examined the patient, made a diagnosis, and logged into the system.

Post Conditions	The patient's treatment plan is saved in the system and becomes visible to

Use Case ID	UC_31 – Submit Treatment Plan
	healthcare staff.
Main Success Scenario	1. The doctor clicks on the “Treatment Plan” tab of the diagnosed patient. 2. Enters details such as treatment duration, medication information, and application frequency. 3. Clicks the “Save and Submit” button. 4. The system saves the treatment plan and shares it with other authorized units.
Alternative Flow	3a. The doctor can use a previously saved template to quickly create a treatment plan.
Exceptional Flow	4a. If mandatory fields are missing, the system displays a warning: “Required fields are not filled” and the record is not completed.
Rules	**R1.** Treatment plans can only be created after the diagnosis. **R2.** Only the doctor can edit the submitted plan.

Use Case ID	UC_32 – Analyze Data & Report Life Support Suggestion
Description	The AI model provides life support suggestions based on the graphical analysis of the patient’s data.
Actors	AI Module
Preconditions	The patient's health data (test results, symptoms, diagnoses, etc.) must be entered into the system, and the graphical analysis module must be active and operational.
Post Conditions	The AI provides recommendations aimed at improving the patient's quality of life.
Main Success Scenario	1. The system collects the patient’s past health data. 2. The graphical analysis module processes the data. 3. The AI model interprets the analysis results. 4. The system generates lifestyle or support plan recommendations. 5. The recommendations are saved in the patient’s history.
Alternative Flow	–
Exceptional Flow	3a. If the required data is missing, the system cannot start the analysis and shows the warning: “Recommendation could not be generated due to insufficient data.”
Rules	**R1.** AI suggestions are generated only if the analysis data is current and complete.

Use Case ID	UC_33 – Analyze Parameters and Report Diagnosis
Description	The AI model analyzes the parameters sent by the doctor and reports the diagnosis.

Use Case ID	UC_33 – Analyze Parameters and Report Diagnosis
Actors	AI Module, Doctor
Preconditions	The doctor must have entered the patient’s parameters (symptoms, test results, etc.) into the system and submitted a diagnosis request. The AI module must be active.
Post Conditions	The AI successfully generates a diagnosis and notifies the doctor through the system.
Main Success Scenario	1. The doctor creates a diagnosis request and sends the required parameters. 2. The system transfers the data to the AI module. 3. The AI model processes and analyzes the data. 4. The AI model produces a diagnosis suggestion. 5. The diagnosis suggestion is delivered to the doctor via the system.
Alternative Flow	3a. The AI model may offer a ranked list of possible diagnoses based on similar past cases.
Exceptional Flow	2a. If parameters are missing or invalid, the AI cannot perform analysis and the system shows the warning: “Diagnosis could not be generated.”
Rules	**R1.** The AI can operate only with pre-validated parameters. **R2.** The AI output is a suggestion; the final decision belongs to the doctor.

Use Case ID	UC_34 – View Doctors & Select
Description	The admin views the list of registered doctors in the system and selects a doctor.
Actors	Admin
Preconditions	The admin must be logged into the system and have permission to access the management panel.
Post Conditions	The selected doctor is identified, and the system navigates to the detail page of the doctor.
Main Success Scenario	1. The admin logs into the system. 2. From the management panel, selects the “Doctor List” tab. 3. The system lists all registered doctors. 4. The admin selects a doctor from the list. 5. The system displays the detail page of the selected doctor.
Alternative Flow	3a. The admin can narrow down search results by filtering (e.g., name, specialty, hospital, etc.).
Exceptional Flow	2a. If the system cannot load the list due to a connection error, it displays the warning: “Doctor list could not be retrieved.”
Rules	**R1.** Only users with admin privileges can access the doctor list. **R2.** The selected doctor's information can only be viewed; editing requires separate authorization.

Use Case ID	UC_35 – View Exams & Select
Description	The admin views the selected doctor's last 12-month examination list and selects an exam.
Actors	Admin
Preconditions	The admin must be logged into the system and must have selected a doctor. Examination data related to the doctor must be recorded in the system.
Post Conditions	A specific examination related to the selected doctor is selected for viewing.
Main Success Scenario	1. The admin selects a doctor from the doctor list. 2. The system lists the last 12 months of examinations for the selected doctor. 3. The admin selects an examination from the list. 4. The system displays the details of the selected examination.
Alternative Flow	2a. The admin can filter the list by date range or patient name to narrow the results.
Exceptional Flow	2b. If there are no examinations in the last 12 months, the system shows the warning: "No records found for this period."
Rules	**R1.** Only users with the Admin role can access the examination history. **R2.** The system lists and filters only data from the last 12 months.

Use Case ID	UC_36 – View Examinations Details
Description	The admin views the details of the selected examination (patient test results, tests, AI and doctor diagnosis).
Actors	Admin
Preconditions	The admin must be logged into the system and must have selected one of the past examinations of the relevant doctor.
Post Conditions	All details related to the selected examination are displayed to the admin.
Main Success Scenario	1. The admin selects one examination from the past examination list. 2. The system loads the examination details. 3. The admin views test results, test data, AI diagnosis suggestions, and the doctor's diagnosis in separate tabs.
Alternative Flow	3a. The admin can filter and view only specific types of data (e.g., only test results).
Exceptional Flow	2a. If any data is missing, the system shows the warning "Relevant examination data is incomplete" and highlights the missing part.
Rules	**R1.** Only admins can access these details. **R2.** The displayed data is read-only and cannot be edited.

Use Case ID	UC_37 – View Treatment Plan
Description	The admin views the treatment plan defined by the doctor for the patient.
Actors	Admin
Preconditions	The admin must have accessed the doctor's examination history and the details of the selected examination.
Post Conditions	The admin successfully views the treatment plan related to the selected patient.
Main Success Scenario	1. The admin selects one of the doctor's past examinations. 2. The system loads the examination details. 3. The admin clicks on the treatment plan tab. 4. The system displays the treatment plan defined by the doctor for the relevant patient.
Alternative Flow	–
Exceptional Flow	2a. If a treatment plan has not yet been created, the system displays the warning: "No treatment plan found for this examination."
Rules	**R1.** The admin only has viewing permission. **R2.** Treatment plans that are missing or not recorded in the system cannot be accessed.

Use Case ID	UC_38 – Check Diagnosis Compatibility
Description	The admin performs a necessity analysis of the patient's treatment plan and enters it into the system.
Actors	Admin
Preconditions	A treatment plan created by the doctor must be recorded in the system. The admin must be able to view this plan.
Post Conditions	The necessity analysis related to the treatment plan is completed and successfully saved in the relevant section of the system.
Main Success Scenario	1. The admin accesses the treatment plan of the relevant patient. 2. Reviews of the items in the treatment plan. 3. Performs an evaluation of the necessity and appropriateness of each item. 4. The system saves the admin's analysis results into the database.
Alternative Flow	3a. The admin may sort the treatment plan items based on their importance.
Exceptional Flow	2a. If the treatment plan is missing or not available in the system, the

Use Case ID	UC_38 – Check Diagnosis Compatibility
	system shows the warning: “No treatment plan found for analysis.”
Rules	**R1.** Only users with the Admin role can perform necessity analysis. **R2.** Analysis data can be revised but not deleted.

Use Case ID	UC_39 – Generate Statistics Based on the Doctor’s Findings
Description	The admin generates diagnostic accuracy statistics of the AI model using feedback provided by the doctor.
Actors	Admin
Preconditions	The doctor must have previously created a diagnosis request and submitted feedback.
Post Conditions	The admin generates diagnostic accuracy statistics for the AI’s performance and saves the analysis results to the system.
Main Success Scenario	1. The doctor submits feedback on the AI diagnosis. 2. The admin analyzes the feedback for training the AI model. 3. The system compares past diagnosis data with the doctor’s actual findings. 4. The system produces statistical data about diagnostic accuracy (e.g., accuracy rate, number of errors). 5. The system displays these results in the admin interface.
Alternative Flow	4a. While comparing past data, the admin may include similar cases to perform a broader analysis.
Exceptional Flow	2a. If the feedback is missing or does not meet the standards, the system cannot start the analysis and displays: “Valid feedback data not found.”
Rules	**R1.** Feedback is mandatory. **R2.** Diagnosis and findings must be in a matchable format.

Use Case ID	UC_40 – Retrain AI Using New Data
Description	The admin retrains the AI model with new data based on the comparison between the doctor and AI diagnoses.
Actors	Admin, AI Module
Preconditions	Past diagnosis data and corresponding doctor feedback must be available in the system.
Post Conditions	The AI model is trained with updated data and provides higher accuracy in new diagnoses.

Main Success Scenario	1. The admin logs into the system. 2. The admin collects the feedback data. 3. The admin formats the data appropriately.

Use Case ID	UC_40 – Retrain AI Using New Data
	4. The admin starts the retraining process for the AI model. 5. The model is successfully trained with the new data and updated in the system. 6. The system displays the message “Model successfully updated.”
Alternative Flow	–
Exceptional Flow	2a. If the feedback data is missing or faulty, the system shows the warning: “Valid data not found, training could not start.”
Rules	**R1.** The new data structure must be compatible with the previously recorded structure. **R2.** Training cannot begin without feedback data.

Use Case ID	UC_42 – Perform Requirement Analysis
Description	The admin analyzes the necessity level of the treatment plans recorded in the system and adds the doctors who prepared low-necessity plans to the investigation list.
Actors	Admin
Preconditions	Treatment plans must be recorded in the system. Requirement analysis data must be available. Doctors must be registered in the system.
Post Conditions	Treatment plans with low necessity level are identified. The relevant doctors are added to the investigation list.
Main Success Scenario	1. The admin logs into the system. 2. Accesses the “Requirement Analysis” module. 3. The system lists the necessity scores of the treatment plans. 4. The admin views those below the threshold value. 5. The admin identifies the treatment plans with low necessity level. 6. Adds the relevant doctor to the “Investigation List.” 7. The system records the action.
Alternative Flow	5a. The admin filters the analysis based on date range or specific doctors. 6a. The admin adds a comment before adding the doctor.
Exceptional Flow	3a. If the necessity score hasn’t been calculated for the treatment plan, the system displays the warning: “Necessity score not available for the related treatment plan.”
Rules	**R1**. A threshold value for the necessity score must be defined. **R2**. The same doctor cannot be added more than once.

Use Case ID	UC_43 – View Investigation List & Select
Description	The admin views the investigation list recorded in the system and selects a doctor to review the details.
Actors	Admin

Preconditions	The investigation list must exist in the system.

Use Case ID	UC_43 – View Investigation List & Select
	At least one doctor must have been previously added to the system.
Post Conditions	The admin is directed to review the details of the selected doctor.
Main Success Scenario	1) The admin logs into the system. 2) Clicks the “Investigation List” tab. 3) The system lists doctors on the active investigation list. 4) The admin selects a doctor from the list. 5) The system displays details of the selected doctor.
Alternative Flow	3a. The admin narrows the list by filtering (date, hospital, branch, etc.). 4a. The admin selects more than one doctor and starts a bulk analysis.
Exceptional Flow	2a. If the investigation list is empty, the system shows the message: “There are currently no doctors in the investigation list.”
Rules	**R1**. No selection can be made if the doctor list is empty. **R2**. Only active records should be listed.

Use Case ID	UC_44 – Examine All Treatment Plans
Description	The admin views the past test results, diagnoses, and treatment plans of the selected doctor within the scope of the investigation list.
Actors	Admin
Preconditions	The admin must be logged into the system. The selected doctor must be on the investigation list. The doctor must have previously created at least one diagnosis and treatment plan for a patient.
Post Conditions	The admin gains information about the doctor’s past activities. The system generates reports containing detailed records.
Main Success Scenario	1) The admin logs into the system. 2) Selects a doctor from the “Examine Doctor” menu. 3) The system lists the past examinations of the selected doctor. 4) The admin selects a record and accesses the details. 5) The system displays the test results, diagnosis suggestions, and treatment plan of the related examination.
Alternative Flow	–
Exceptional Flow	3a. If there is no past record for the doctor, the system displays the warning: “No past examination record found for this doctor.”
Rules	**R1**. There must be at least one past record. **R2**. The admin only has viewing privileges.

Use Case ID	UC_45 – Record as Malicious

Description	The admin analyzes the necessity rates of past treatment plans and if a plan is identified as unnecessary above a certain threshold, the system marks the

Use Case ID	UC_45 – Record as Malicious
	doctor with the “malicious” label.
Actors	Admin
Preconditions	The admin must be logged into the system. The doctor’s past treatment plans must be recorded in the system. Necessity analysis data must exist in the system.
Post Conditions	The system adds the “malicious” label to the doctor. The doctor is added to the monitoring list.
Main Success Scenario	1. The admin logs into the system. 2. The admin reviews the doctor’s past treatment plans. 3. The system calculates the unnecessary treatment ratio. 4. If the ratio is below 70%, the system marks the doctor as “Malicious.” 5. The system adds the doctor to the monitoring list and issues a warning.
Alternative Flow	4a. If the necessity ratio is above 70%, the process ends without marking the doctor.
Exceptional Flow	3a. If necessity analysis data is missing, the system shows the warning: “Insufficient analysis data.”
Rules	**R1.** Necessity ratio must be less than or equal to 70%. **R2.** If necessity data is missing, no marking can be done.

Use Case ID	UC_46 – Notify the Case to the Ministry of Health
Description	The admin sends the diagnosis and treatment plan applied by a doctor labeled as *malicious*, along with the related test results, to the Inspection Board of the Ministry of Health.
Actors	Admin
Preconditions	The admin must be logged into the system. The diagnosis, treatment plan, and test results to be sent must be complete in the system. The doctor must have the *malicious* label.
Post Conditions	The relevant information is successfully submitted to the Ministry of Health’s Inspection Board. A “sent for inspection” tag is generated in the system.
Main Success Scenario	1. The admin logs into the system. 2. The admin views the diagnosis and treatment details of the doctor labeled as *malicious*. 3. The admin checks the treatment plan and test results. 4. The admin sends all information to the Ministry of Health’s Inspection Board with one click. 5. The system confirms successful delivery and creates a record.
Alternative Flow	–
Exceptional Flow	3a. If the treatment plan or test result is missing, the system does not allow submission and displays the error: “Submission cannot proceed due to missing information.”
Rules	–

Use Case ID	UC_47 – View Patient Test List
Description	The hospital laboratory views the list of test requests submitted for patients in the system.
Actors	Hospital Laboratory Staff
Preconditions	The laboratory staff must be logged into the system. At least one test request must have been submitted to the system by doctors or hospital administration.
Post Conditions	Test requests are successfully displayed. If needed, details can be accessed by selecting an item from the list.
Main Success Scenario	1. The lab staff logs into the system. 2. Clicks on the "View Patient Test List" tab. 3. The system lists all test requests in chronological order. 4. The actor can view the details of each test request.
Alternative Flow	3a. The actor applies filters based on date or test type, and the list updates accordingly.
Exceptional Flow	2a. If there are no test requests in the system, it shows the message: “No test requests found.”
Rules	**R1**. The patient with a test request must have an associated examination record.

Use Case ID	UC_48 – Select Patient for Testing
Description	The laboratory staff selects the next patient in the queue from the list and initiates the testing process.
Actors	Hospital Laboratory Staff
Preconditions	The lab staff must be logged into the system. There must be at least one patient waiting for testing in the system.
Post Conditions	The test process is initiated for the selected patient. The test status is updated to **“In process”**.
Main Success Scenario	1. Lab staff logs into the system. 2. Opens the "Test Requests" list. 3. Selects the next patient in the queue. 4. Clicks the “Start Test” button. 5. The system initiates the test and updates the patient's status to **“In process”**.
Alternative Flow	–
Exceptional Flow	2a. If there is no patient waiting for testing, the system shows the message: “No patient waiting for testing.” 4a. If the user clicks “Start Test” without selecting a patient, the system shows the warning: “Please select a patient.”
Rules	**R1**. The test cannot be started without selecting a patient. **R2**. The test cannot be started for a patient without a test request.

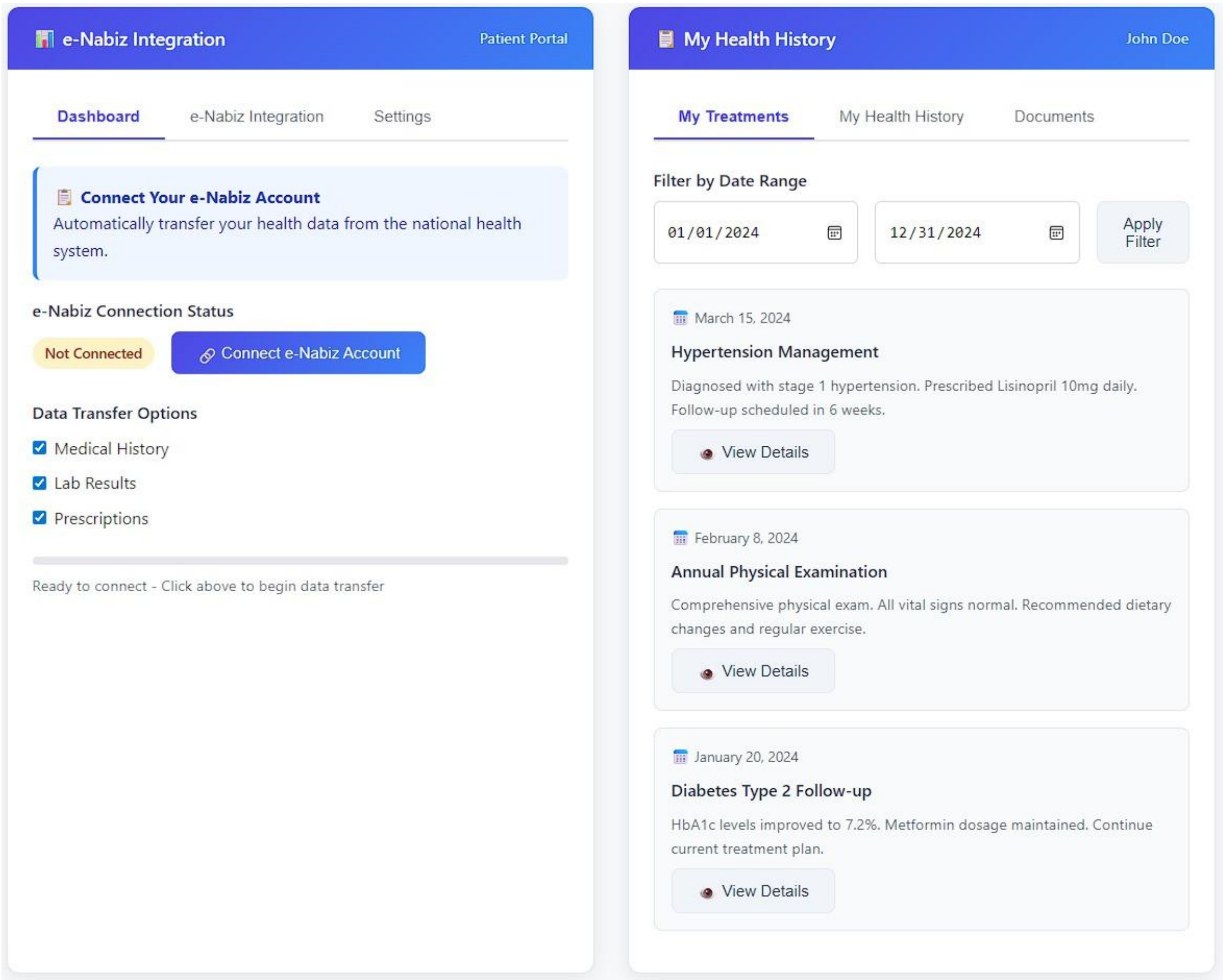
Use Case ID	**UC-49 – Upload Test Results**
Description	The laboratory staff uploads the results of completed patient tests into the system.
Actors	Hospital Laboratory Staff
Preconditions	Test procedures must be completed.
Post Conditions	Test results are added to the patient's records in the system. The doctor and relevant personnel can view the test results.
Main Success Scenario	1. Lab staff logs into the system. 2. Navigates to the “Upload Test Result” page. 3. Selects the patient and type of test. 4. Uploads the test result to the system (either file or manual entry). 5. The system saves the result and grants access to the doctor.
Alternative Flow	4a. When uploading a file, the system automatically displays the results and summary info. 4b. If the user prefers manual entry, they fill in the results using a form.
Exceptional Flow	3a. If an attempt is made to upload without selecting a patient, the system shows: “Please select a patient.” 4c. If the file format is invalid, the system displays an error message.
Rules	**R1**. Test results cannot be uploaded without selecting a patient. **R2**. Test result file must be in a valid format.

Use Case ID	**UC_50 – View Case Details**
Description	The Inspection Board reviews case details related to doctors flagged as *malicious*.
Actors	Inspection Board Officer
Preconditions	The Inspection Board Officer must be logged into the system. There must be at least one doctor flagged as *malicious* in the system. The doctor must have at least one registered case (examination record) in the system.
Post Conditions	The officer gains access to the doctor's test results, diagnosis, treatment plan, and AI-based diagnosis along with all other case details.
Main Success Scenario	1. The Inspection Board Officer logs into the system. 2. Selects a doctor from the *Malicious Doctors* list. 3. The list of cases for the selected doctor is displayed. 4. A case is clicked. 5. The system displays all details including test results, diagnoses, treatment plans, AI recommendations, and any comments if available.
Alternative Flow	3a. Cases can be filtered (by date range, type, etc.). 4a. The officer can compare multiple cases of a doctor simultaneously.
Exceptional Flow	3b. If the selected doctor has no registered case data, the system displays: “No registered case found.” 5a. If case details fail to load, the system shows an error message.

Use Case ID	UC_50 – View Case Details
Rules	**R1**. The Inspection Board can only view doctors who have been flagged. **R2**. If another user tries to access, the system displays: "Access Denied."

Use Case ID	UC_51 – Suspend Doctor Access
Description	The Inspection Board suspends the activity of a doctor in the system who has been decided to be prosecuted due to malicious behavior.
Actors	Inspection Board Officer
Preconditions	The Inspection Board Officer must be logged into the system. The doctor must already be flagged as *malicious*. The relevant case details must have been reviewed.
Post Conditions	The doctor's access to the system is restricted. All active operations related to the doctor are suspended.
Main Success Scenario	1. The Inspection Board Officer logs into the system. 2. Opens the list of doctors flagged as *malicious*. 3. After reviewing the details, clicks the "Suspend Operations" option. 4. The system disables the doctor's login rights and action permissions. 5. The doctor is marked as an inactive user. 6. This action is recorded as an audit log in the system.
Alternative Flow	4a. Before suspending, the system shows a confirmation message: "Are you sure?" 4b. If the board cancels the action, no change is made.
Exceptional Flow	3a. If the doctor is already inactive, the system shows: "This doctor is already suspended." 5a. If there is a permission error, the system shows an error and does not proceed.
Rules	**R1**. A suspended doctor cannot log into the system.

3.2. User interfaces

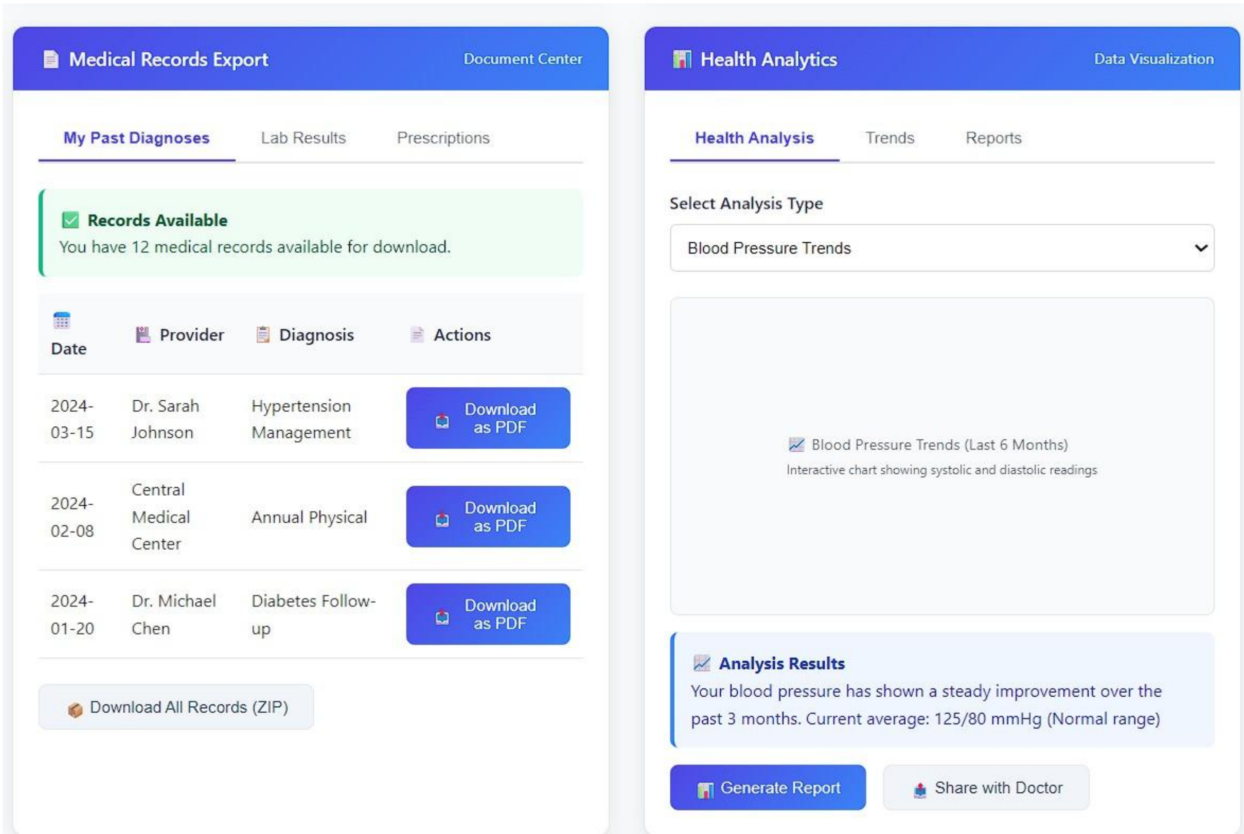


- **e-Nabiz Integration**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Connect e-Nabiz Account Button	Initiate data transfer from national health system	User click input	Triggers authentication and data transfer
Data Transfer Options (Checkboxes)	Allow user to select which data types to transfer	User checkbox selections	Determines scope of data integration

- **My Health History**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Filter by Date Range	Enable filtering of diagnosis and treatment records	User date input	Controls which records are displayed
Medical Record Cards	Display previous diagnosis and treatment plans	Patient medical database	Shows filtered treatment history
Treatment Title	Show specific diagnosis names	Medical record data	Links to detailed treatment information
View Details Button	Access complete diagnosis and treatment details	User click input	Opens detailed view for UC_10

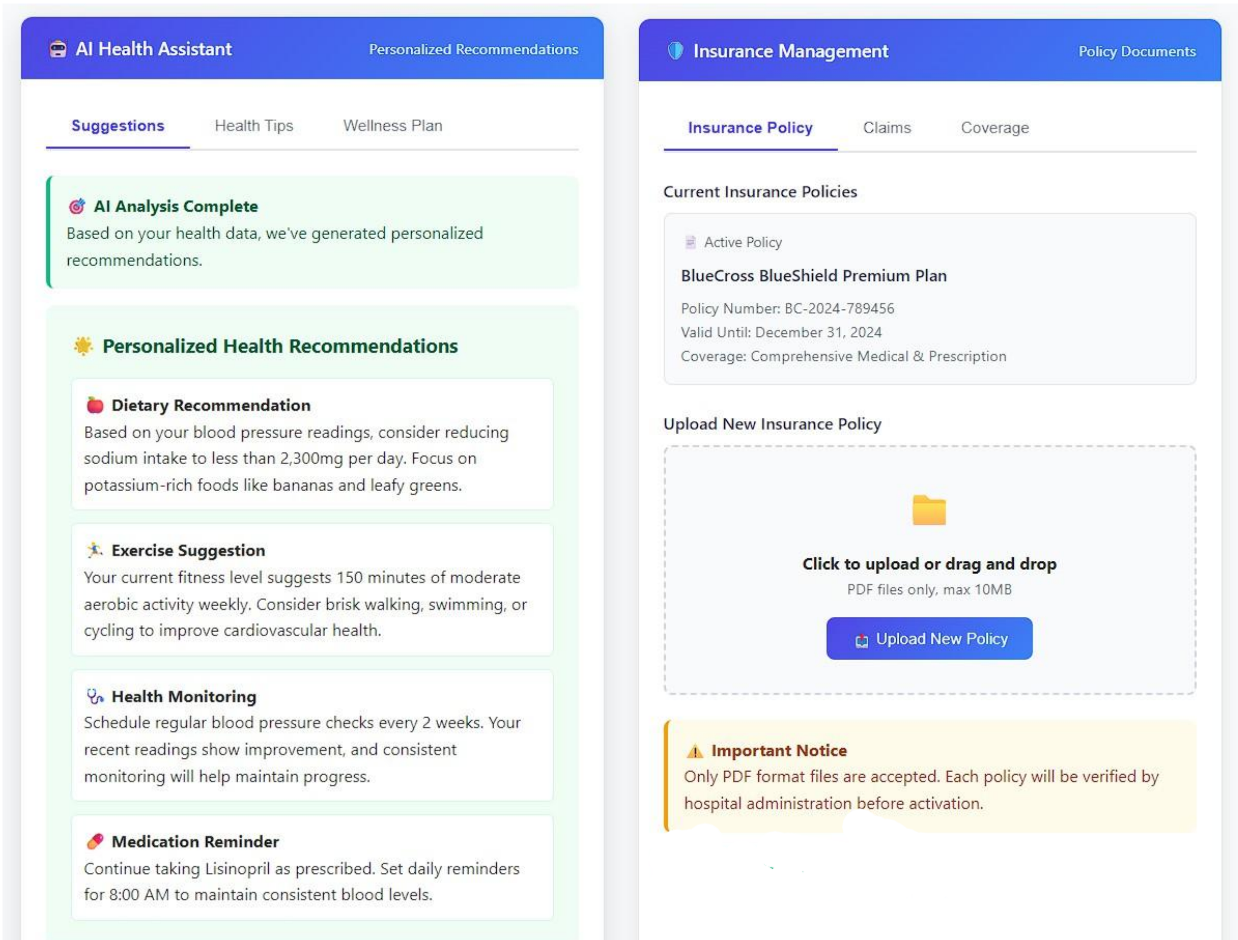


- **Medical Records Export**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
My Past Diagnoses Tab	Navigate to diagnosis records for PDF export	User click input	Primary tab for UC_11 functionality
Medical Records Table	Display patient's past diagnoses and treatment plans	Patient medical database	Lists records available for PDF export
Provider Column	Show healthcare provider names	Doctor/hospital database	Links records to healthcare providers
Download as PDF Button	Generate and download individual record as PDF	User click input	Triggers PDF generation for specific record

- **Health Analytics**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Health Analysis Tab	Navigate to graphical analysis section	User click input	Primary tab for UC_12 functionality
Select Analysis Type Dropdown	Choose type of health data visualization	User selection input	Determines what data gets analyzed
Chart Visualization Area	Display graphical analysis of health data	Patient health data processing	Shows trends and patterns visually
Analysis Results Section	Show interpreted results of health data	AI analysis of patient data	Provides insights from graphical analysis
Generate Report Button	Create detailed analysis report	User click input	Produces comprehensive health analysis

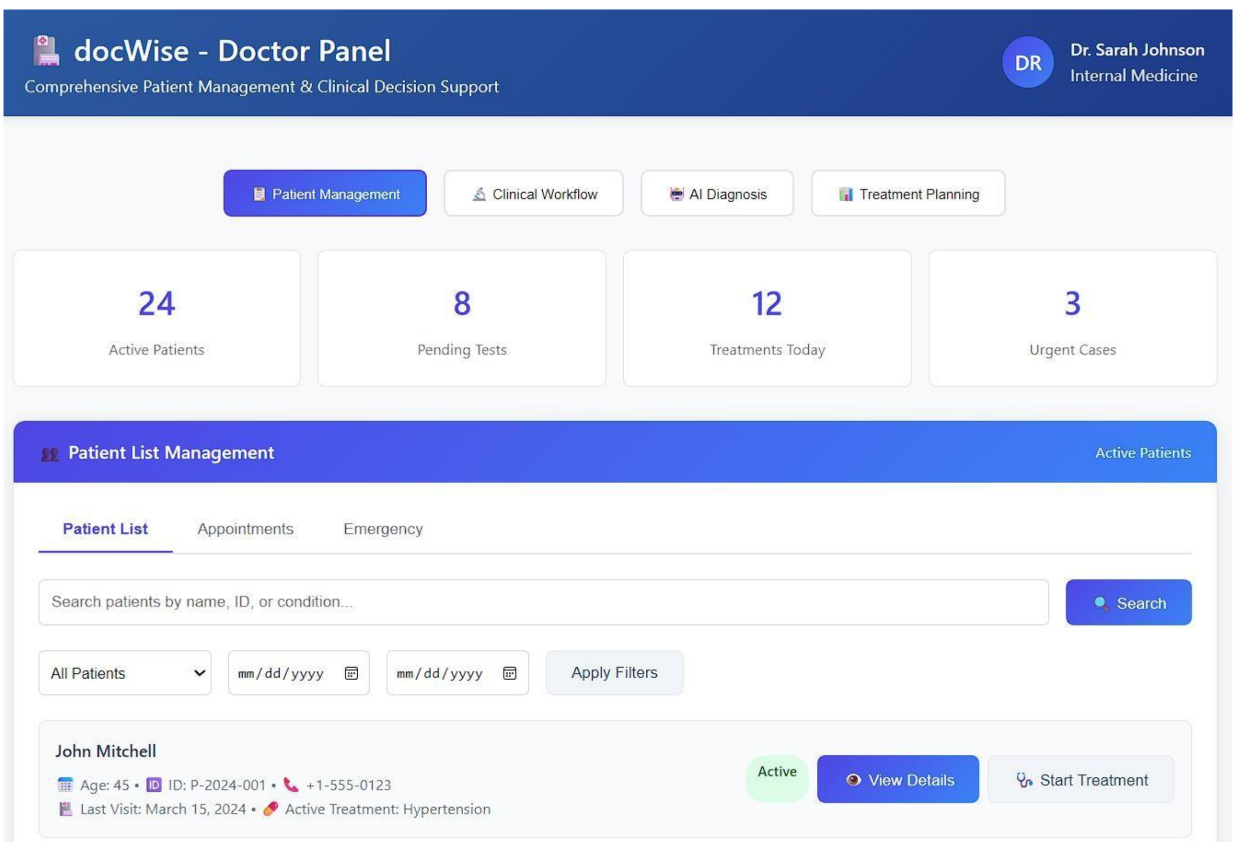


- **AI Health Assistant**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Suggestions Tab	Navigate to AI-generated recommendations	User click input	Primary tab for UC_13 functionality
Personalized Health Recommendations Section	Display AI-generated life support suggestions	AI module analysis output	Contains all AI-generated recommendations

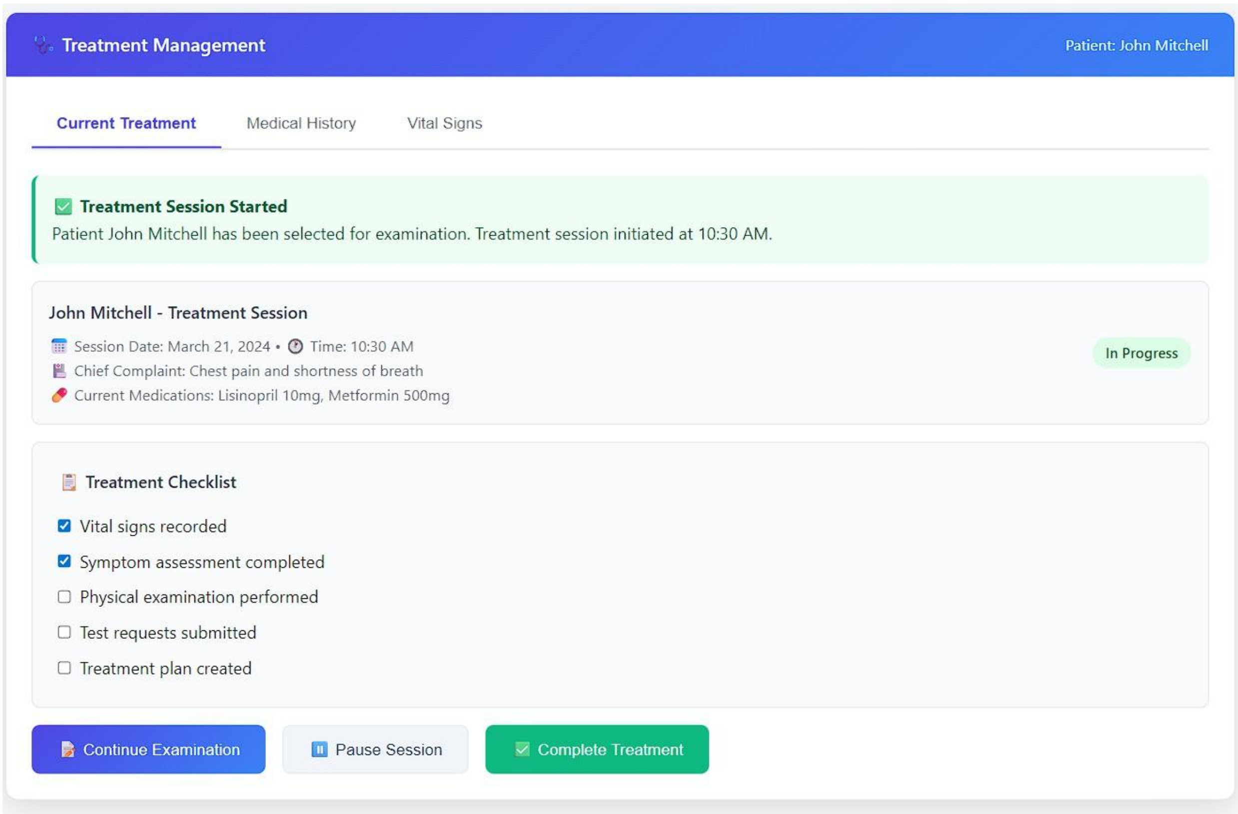
- **Insurance Management**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Insurance Policy Tab	Navigate to policy management section	User click input	Primary tab for UC_14 functionality
Current Insurance Policies Section	Display existing policy information	Insurance database	Shows currently active policies
Upload New Insurance Policy Section	Enable new policy document upload	User file input	Core functionality for UC_14



- **Patient Management**

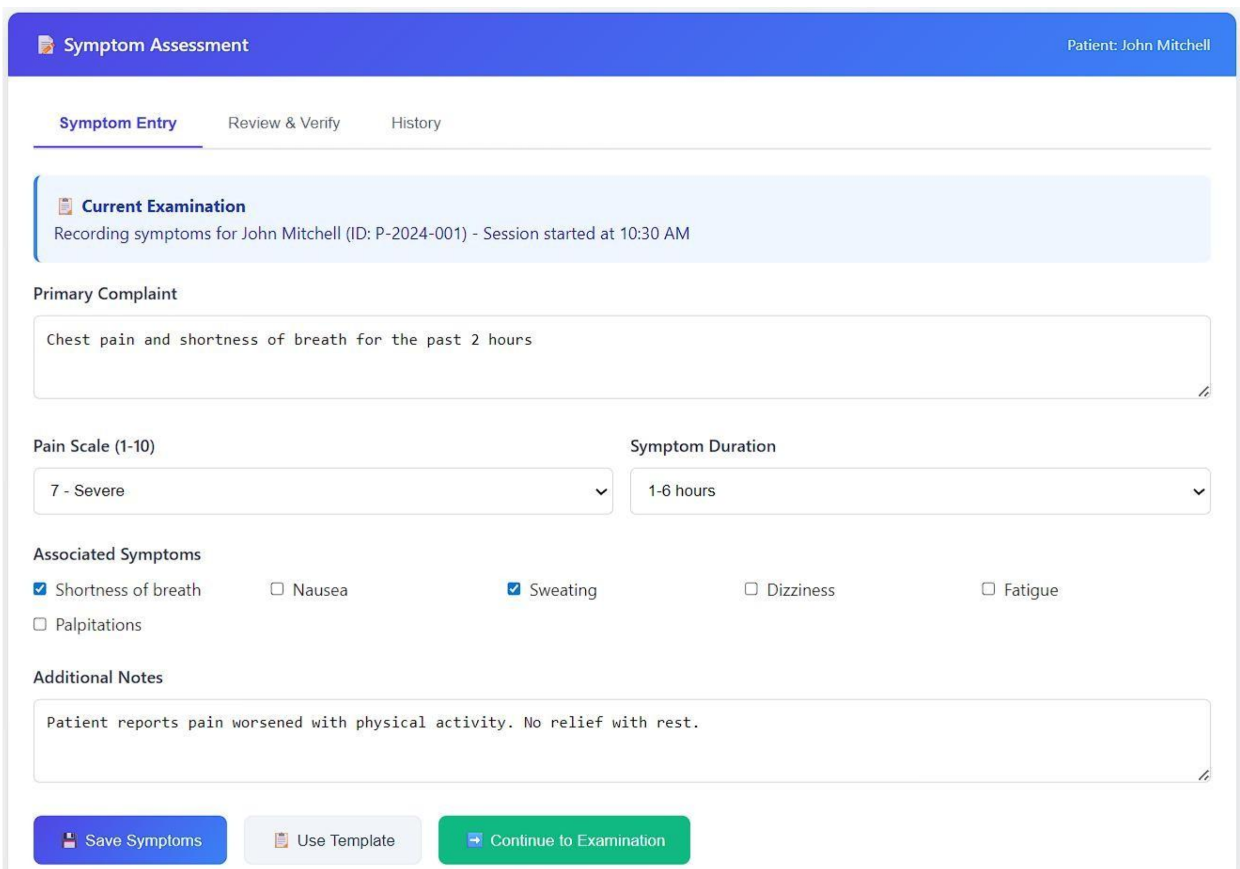
Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Patient Management Tab	Navigate to patient list functionality	User click input	Primary navigation for UC_22
Patient Search Bar	Search patients by name, ID, or condition	User text input	Filters patient list for UC_22
Apply Filters Button	Execute selected filtering criteria	User click input	Updates patient list based on filters
Patient Card (John Mitchell)	Display individual patient information	Patient database	Shows patient details for selection in UC_22
Start Treatment Button	Initiate treatment process for selected patient	User click input	Core functionality for UC_23 - Start Treatment Process



- **Treatment Management**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Treatment Management Header	Display active treatment session details	Treatment session system	Shows selected patient for UC_23
Current Medications Information	Show patient's active medication list	Patient medication database	Context for treatment planning
Treatment Checklist	Track treatment process completion steps	Treatment workflow system	Monitors UC_23 process progression
Continue Examination Button	Proceed with next treatment steps	User click input	Advances UC_23 treatment process

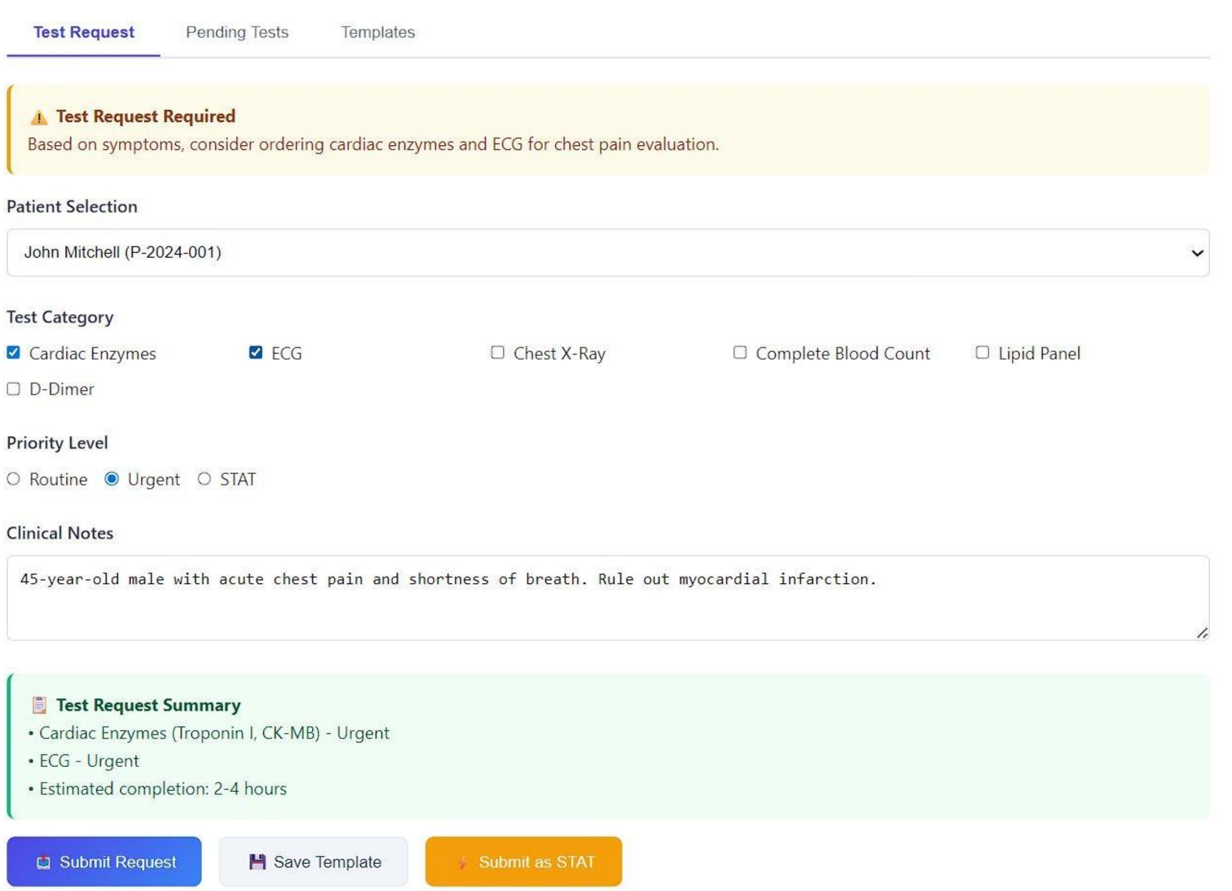
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Pause Session Button	Temporarily halt treatment session	User click input	Allows UC_23 session management
Complete Treatment Button	Finalize treatment session	User click input	Completes UC_23 treatment process



- **Symptom Assessment**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs

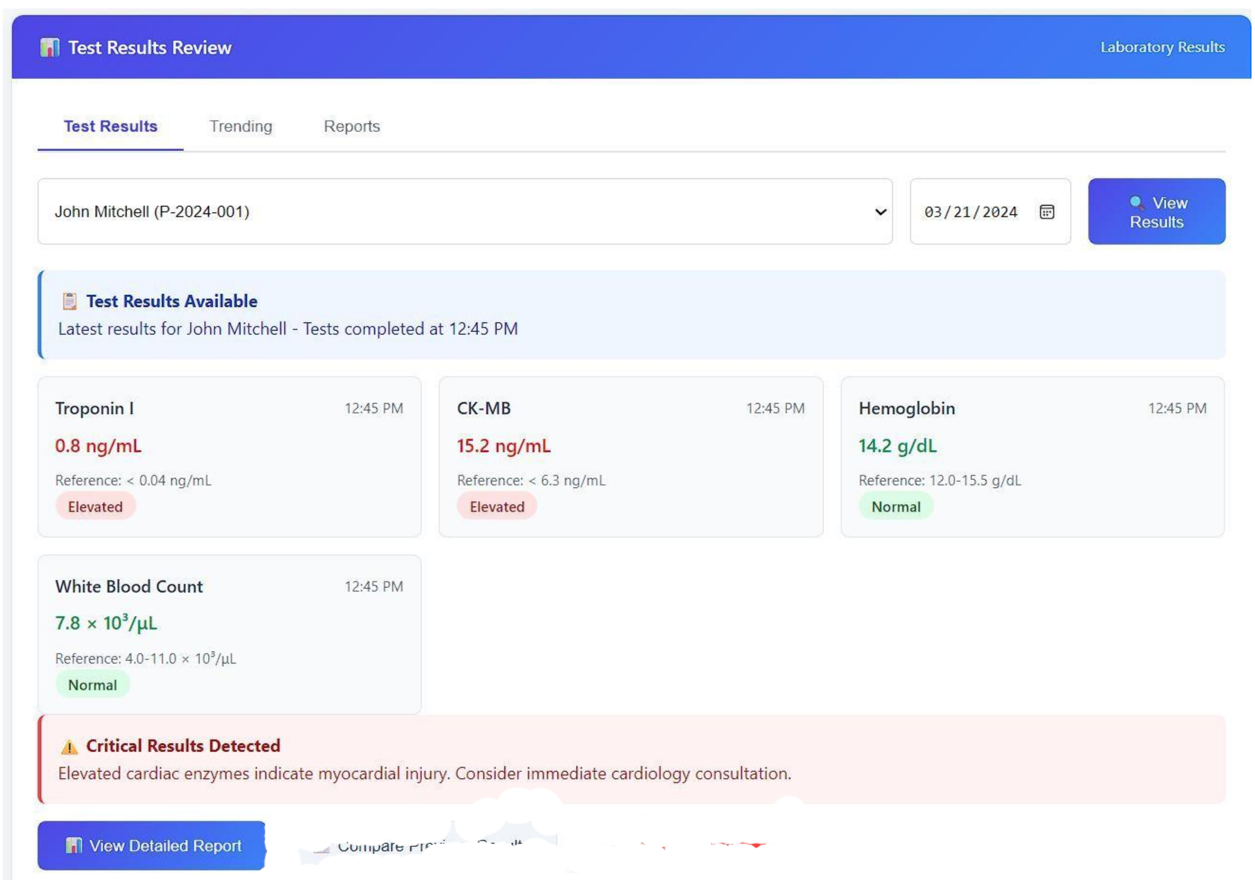
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Symptom Entry Tab	Navigate to symptom input functionality	User click input	Primary tab for UC_24 - Input Patient Symptoms
Primary Complaint Text Area	Enter patient's main symptom description	Doctor text input	Core input for UC_24 symptom recording
Pain Scale Dropdown	Select pain intensity level (1-10)	Doctor selection input	Quantifies symptom severity for UC_24
Symptom Duration Dropdown	Select how long symptoms have persisted	Doctor selection input	Temporal context for UC_24 symptom data
Associated Symptoms Checkboxes	Select additional symptoms present	Doctor checkbox selections	Comprehensive symptom collection for UC_24
Additional Notes Text Area	Enter supplementary symptom observations	Doctor text input	Additional symptom context for UC_24
Save Symptoms Button	Store symptom data in patient record	User click input	Core UC_24 action - saves symptom information
Use Template Button	Apply predefined symptom templates	User click input	Efficiency feature for UC_24 symptom entry
Continue to Examination Button	Proceed to next examination phase	User click input	Advances to post-UC_24 workflow steps



- **Laboratory Test Management**

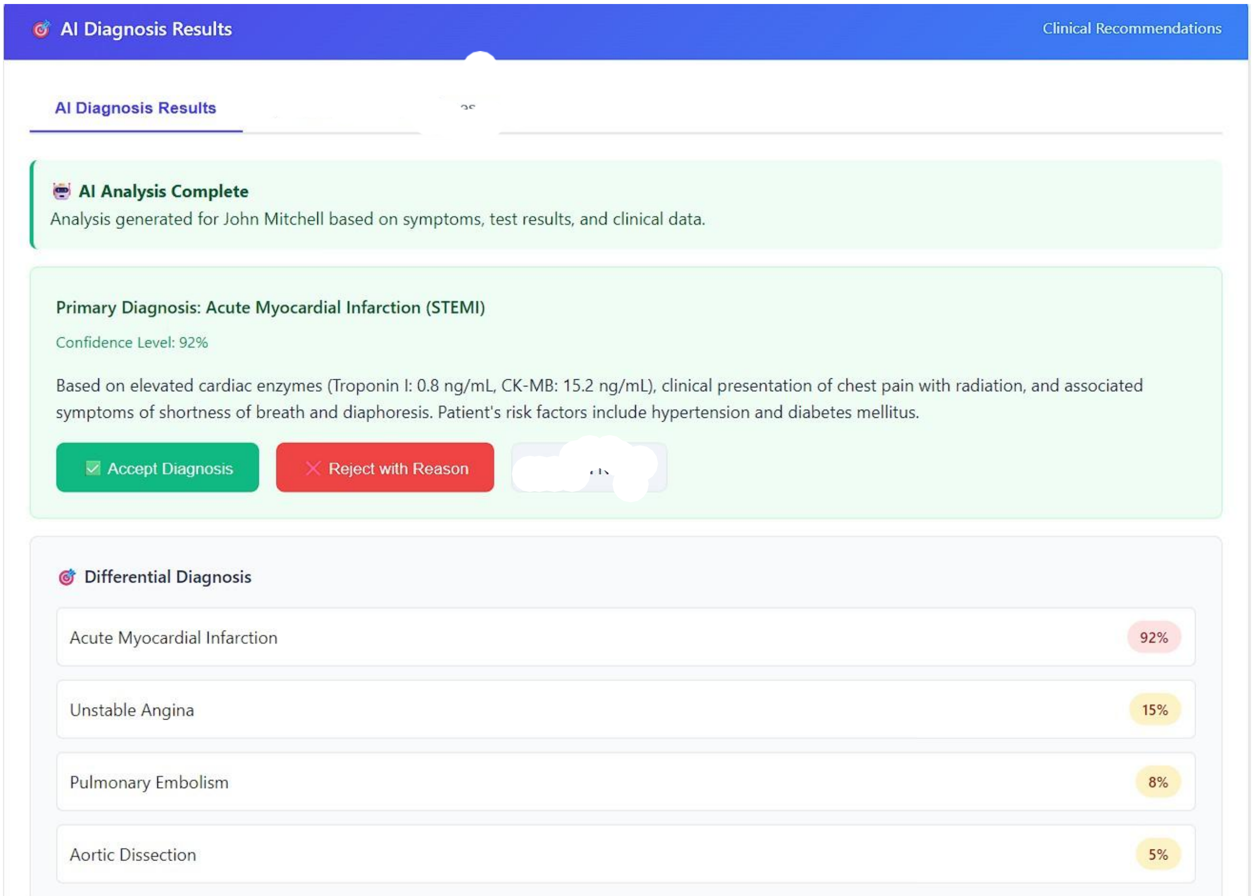
Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Test Category Checkboxes	Select specific laboratory tests to order	Doctor checkbox selections	Core functionality for UC_25 test ordering
Priority Level Radio Buttons	Set urgency level for test processing	Doctor selection input	Determines test processing priority for UC_25
Clinical Notes Text Area	Provide clinical justification for tests	Doctor text input	Clinical indication required for UC_25
Test Request Summary	Display ordered tests and priority	System-generated summary	Confirms UC_25 test request details

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Submit Request Button	Send test orders to hospital management	User click input	Core UC_25 action - submits test requests
Save Template Button	Save test combination for future use	User click input	Efficiency feature for UC_25
Submit as STAT Button	Submit with highest priority level	User click input	Emergency submission option for UC_25



- **Test Results Review**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Patient Selection Dropdown	Select patient to view test results	Doctor selection input	Identifies patient for UC_26 test viewing
View Results Button	Access test results for selected patient/date	User click input	Core action for UC_26 - retrieves test results
Test Results Cards Grid	Display individual test results	Laboratory database	Shows test results data for UC_26
Test Value Display	Show numerical test results	Laboratory test data	Core data element for UC_26
Reference Range Display	Show normal value ranges	Laboratory reference data	Comparison context for UC_26 results
Test Status Indicators	Show if results are normal/elevated	Laboratory analysis	Interpretation support for UC_26
Critical Results Alert	Highlight abnormal test findings	Laboratory flagging system	Clinical decision support for UC_26
View Detailed Report Button	Access comprehensive test analysis	User click input	Extended view option for UC_26

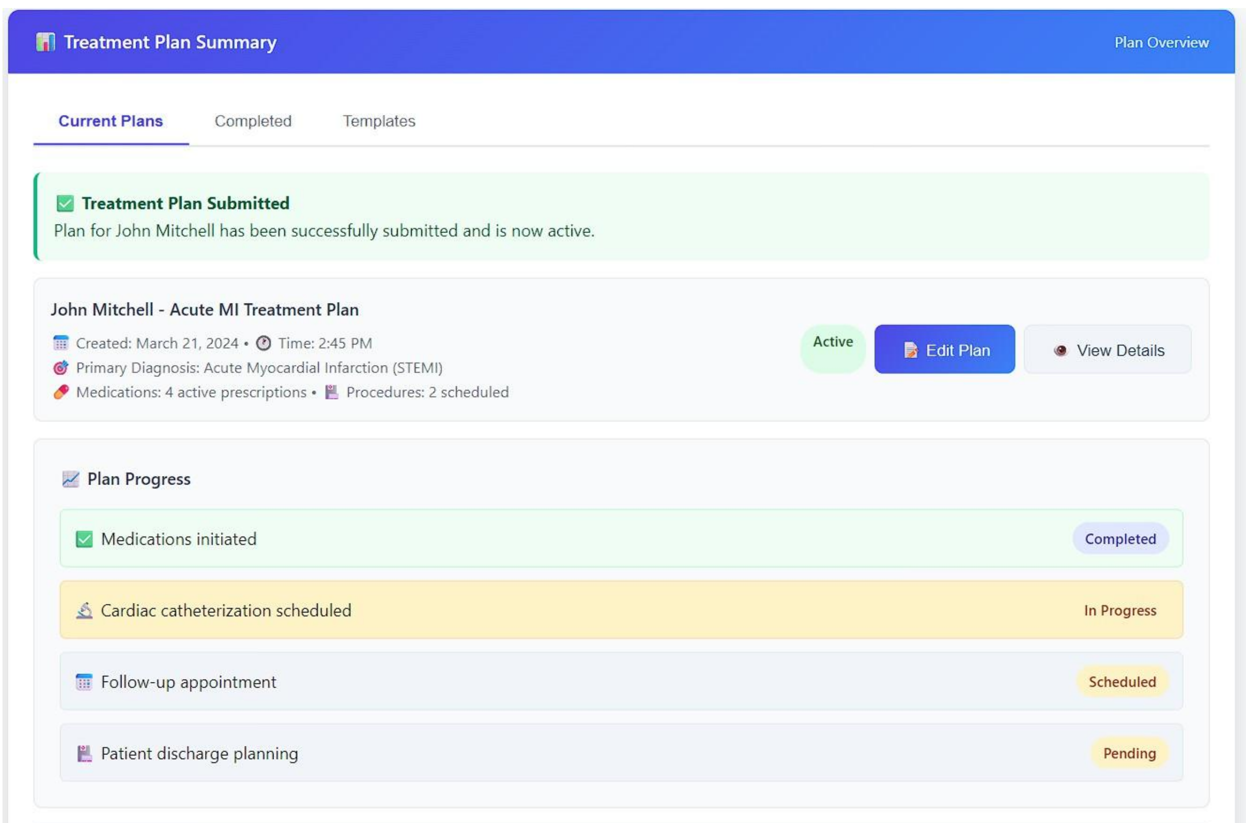


- **AI Diagnosis Results**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
AI Diagnosis Results Header	Display AI-generated diagnosis module	AI analysis system	Shows context for UC_27-30 functionality
Primary Diagnosis Card	Display main AI-generated diagnosis	AI analysis output	Core result of UC_27 - Request AI Diagnosis
Confidence Level Display	Show AI confidence percentage	AI diagnostic algorithm	Reliability indicator for UC_27 results
Clinical Description Text	Provide detailed diagnosis explanation	AI analysis reasoning	Justification for UC_27 AI diagnosis

Accept Diagnosis	Approve AI diagnosis	User click input	Core action for UC_29 -

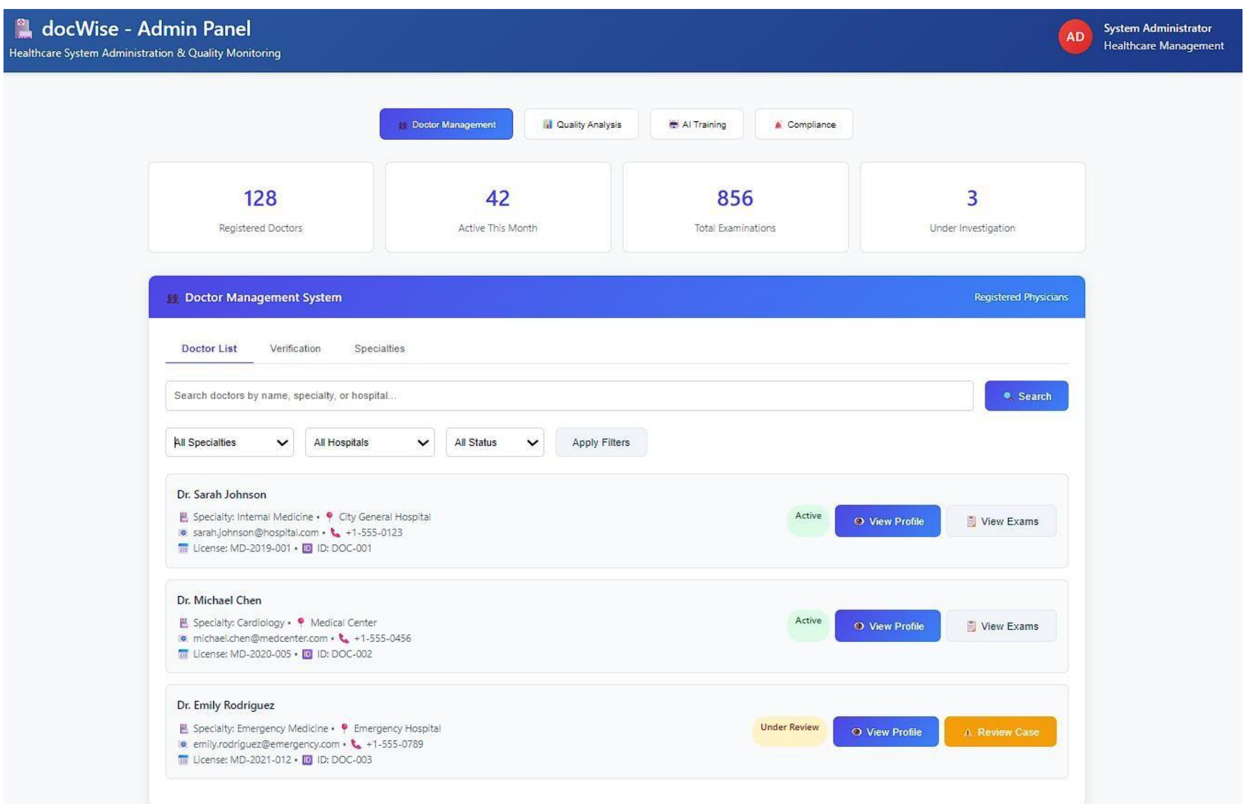
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Button	suggestion		Doctor Accepts Diagnosis
Reject with Reason Button	Decline AI diagnosis with explanation	User click input	Core action for UC_30 - Submit Rejection Reason
Differential Diagnosis Section	Display alternative diagnosis options	AI analysis output	Extended results from UC_27 AI request



- **Treatment Plan Summary**

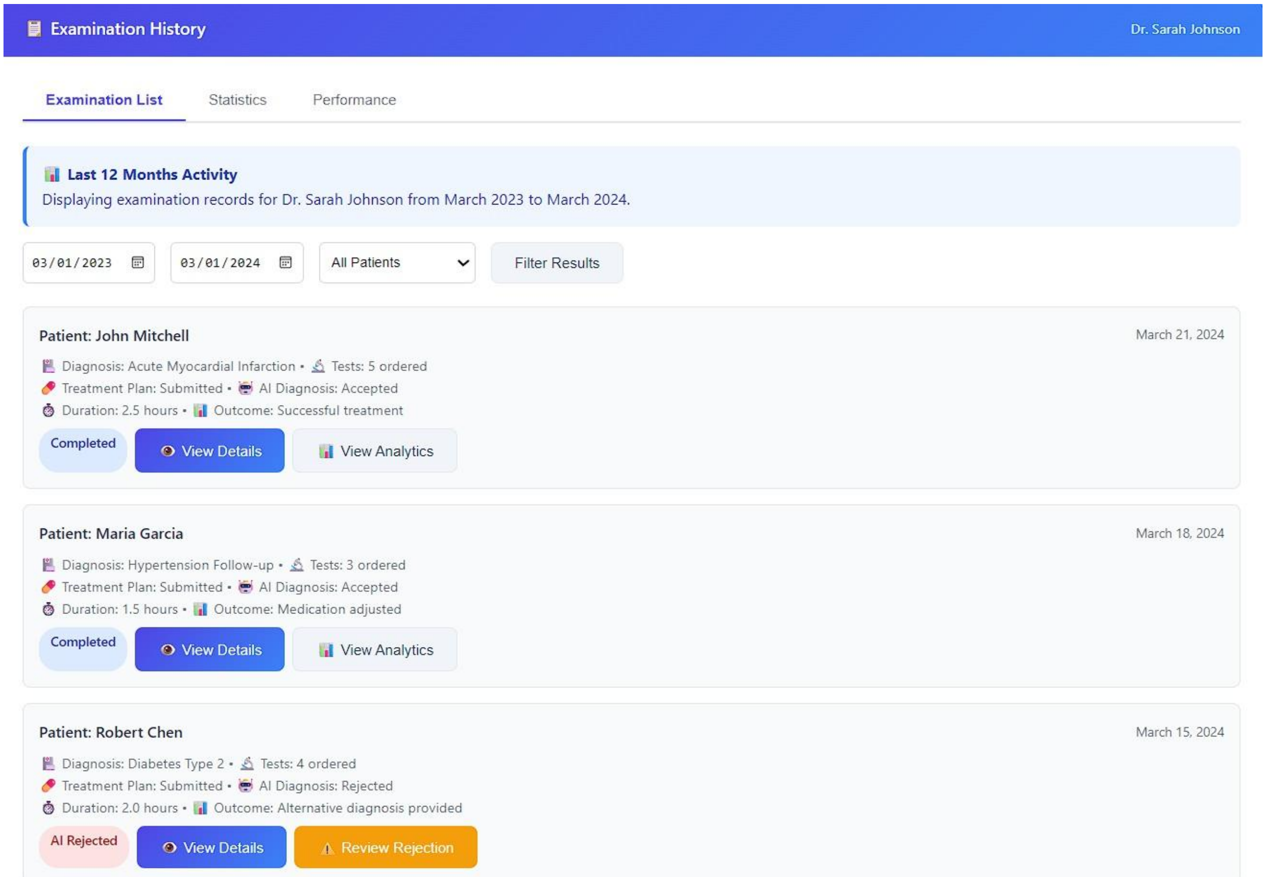
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Treatment Plan Summary Header	Display treatment plan management module	Treatment planning system	Shows context for UC_31 functionality
Treatment Plan Card	Display submitted treatment plan details	Treatment plan database	Shows result of UC_31 submission
Edit Plan Button	Modify existing treatment plan	User click input	Allows UC_31 plan modifications
Plan Progress Section	Track treatment plan implementation	Treatment execution system	Shows UC_31 plan implementation status



Admin Panel - Doctor Management - UC_34, UC_35

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Doctor Management Tab	Navigate to doctor management functionality	User click input	Primary tab for UC_34 - View Doctors & Select
Doctor List Tab	Access list of registered doctors	User click input	Core tab for UC_34 - View Doctors & Select
Search Button	Execute doctor search query	User click input	Triggers doctor list filtering for UC_34
Doctor Information Cards	Display individual doctor details	Doctor database	Shows doctor information for UC_34 selection
View Exams Button	Access doctor's examination history	User click input	Core functionality for UC_35 - View Exams & Select
Review Case Button	Access case review for flagged doctors	User click input	Quality control action for flagged doctors

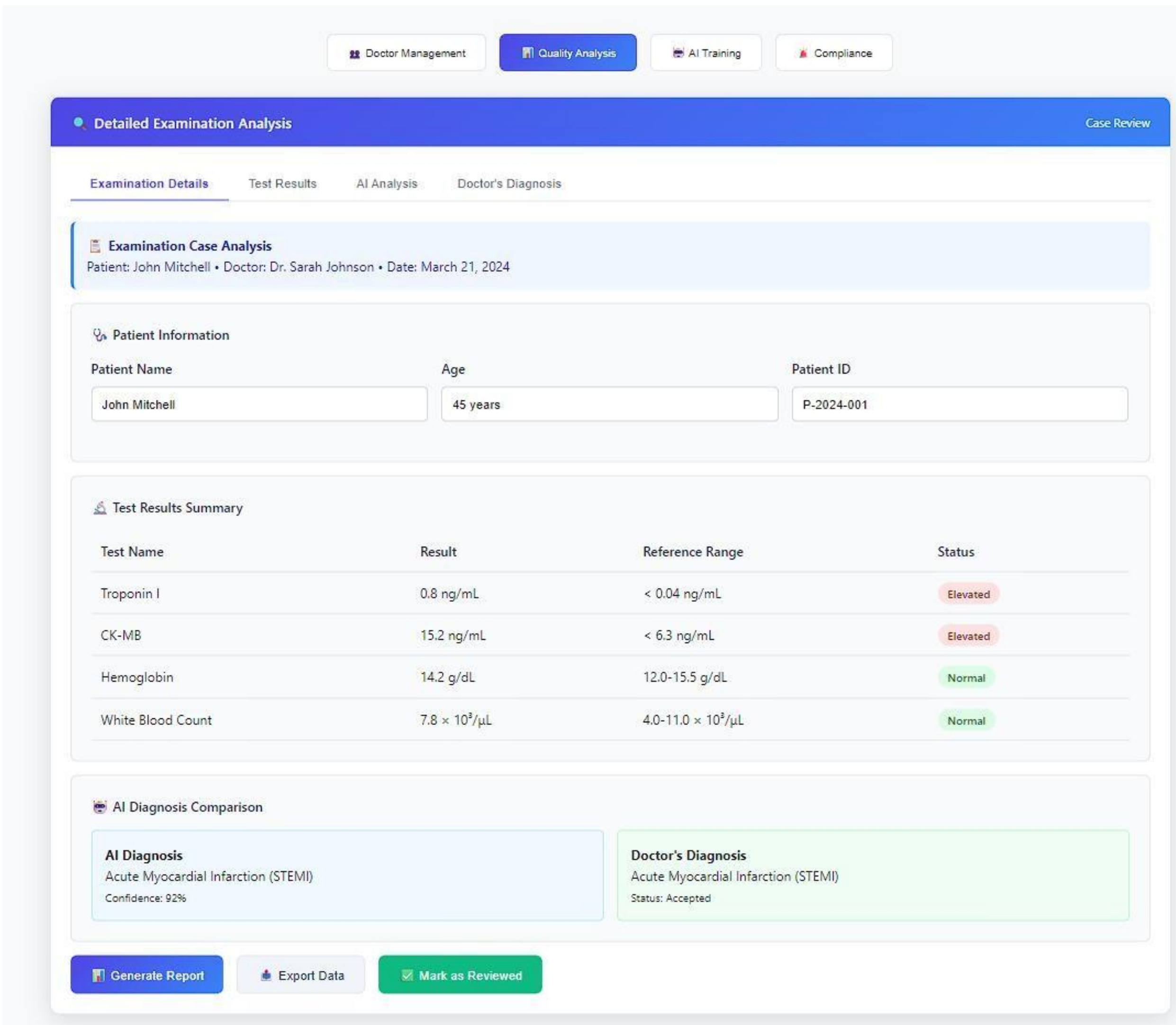


- **Examination History**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Examination List Tab	Navigate to examination history viewing	User click input	Primary tab for UC_35 - View Exams & Select
Patient Filter Dropdown	Filter examinations by patient type	Admin dropdown selection	Refines examination list for UC_35
Examination Record Cards	Display individual examination details	Examination database	Shows examination data for UC_35 selection
View Details Button	Access complete examination information	User click input	Detailed view functionality for UC_35

View Analytics Button	Access examination performance metrics	User click input	Analysis feature for UC_35 examinations

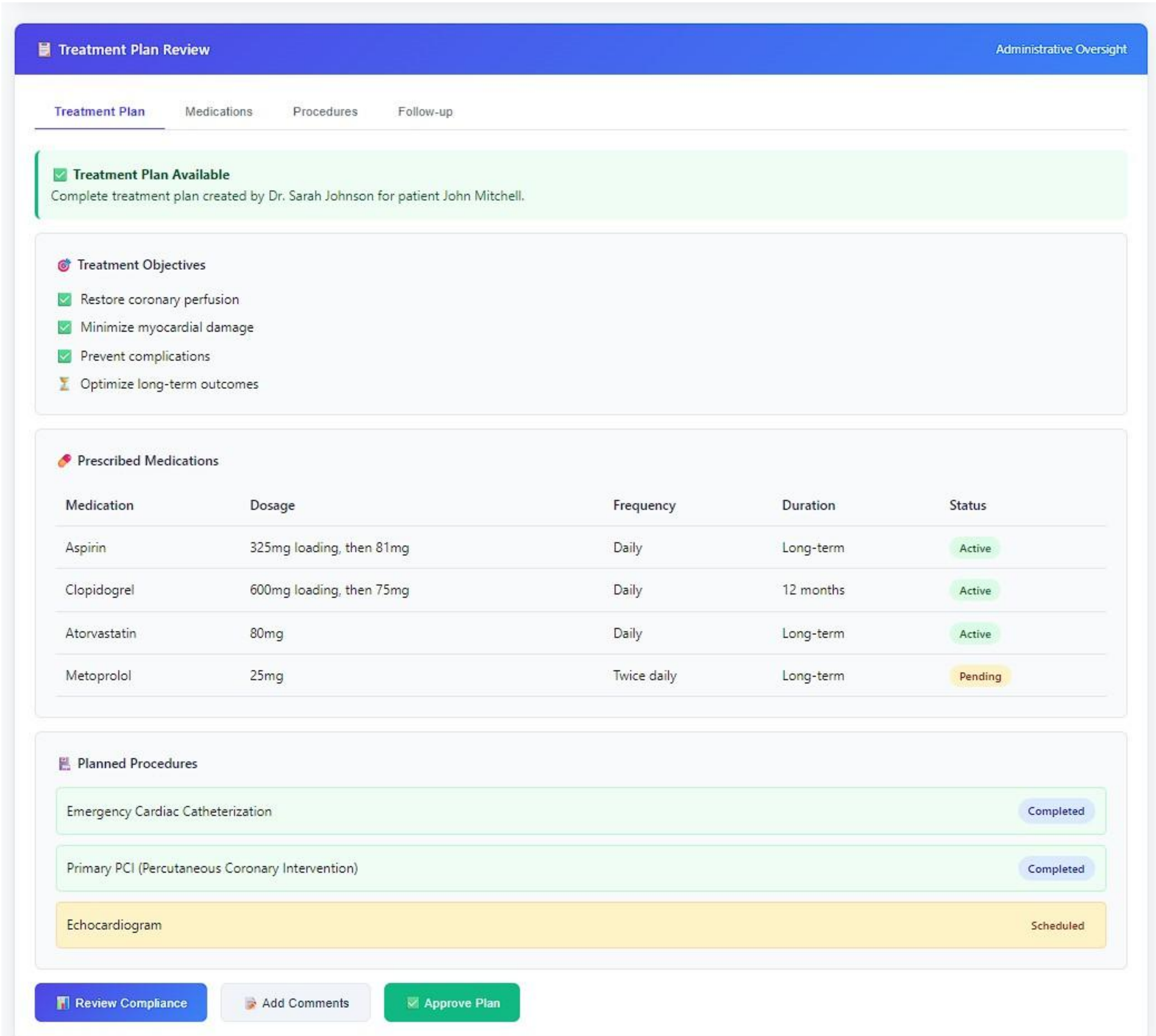
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Review Rejection Button	Access AI rejection review details	User click input	Quality control feature for flagged examinations



- **Detailed Examination Analysis**

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs

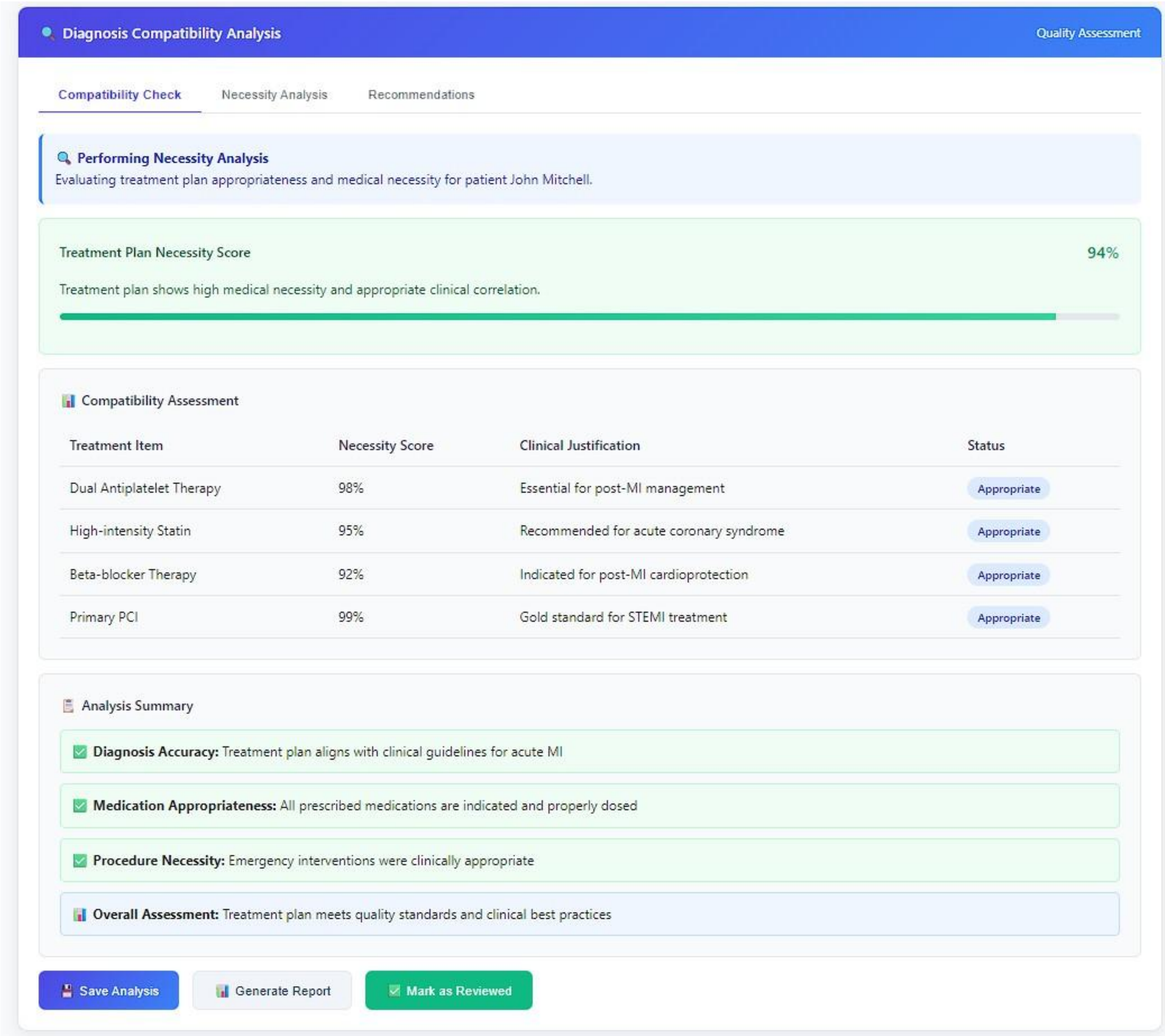
Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Examination Details Tab	Access detailed examination information	User click input	Primary tab for UC_36 - View Examinations Details
Test Results Tab	Access laboratory test results	User click input	Secondary view for UC_36 examination data
AI Analysis Tab	Access AI diagnosis information	User click input	AI component view for UC_36 examination
Doctor's Diagnosis Tab	Access doctor's diagnosis details	User click input	Doctor component view for UC_36 examination
Patient Information Section	Display patient demographics	Patient database	Shows patient details for UC_36 examination
Test Results Summary Section	Display laboratory test outcomes	Laboratory database	Shows test data for UC_36 examination
AI Diagnosis Comparison Section	Compare AI and doctor diagnoses	AI system and doctor input	Shows diagnostic correlation for UC_36
Generate Report Button	Create comprehensive examination report	User click input	Report generation for UC_36
Export Data Button	Export examination data	User click input	Data export feature for UC_36
Mark as Reviewed Button	Indicate examination review completion	User click input	Review completion action for UC_36



- **Treatment Plan Review**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Treatment Plan Tab	Navigate to treatment plan viewing	User click input	Primary tab for UC_37 - View Treatment Plan
Medications Tab	Access medication details	User click input	Medication component view for UC_37
Procedures Tab	Access procedure details	User click input	Procedure component view for UC_37

Name of item	Description of purpose	Source of input or destination of output	Relationships to other inputs/outputs
Follow-up Tab	Access follow-up plan details	User click input	Follow-up component view for UC_37
Medications Table	Present medications in tabular format	Treatment plan database	Organized medication data for UC_37
Planned Procedures Section	Display scheduled medical procedures	Treatment plan database	Shows procedure details for UC_37
Review Compliance Button	Assess treatment plan compliance	User click input	Quality review action for UC_37
Add Comments Button	Include administrative notes	User click input	Commentary feature for UC_37
Approve Plan Button	Administratively approve treatment plan	User click input	Administrative approval for UC_37



- **Diagnosis Compatibility Analysis**

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Compatibility Check Tab	Navigate to diagnosis compatibility analysis	User click input	Primary tab for UC_38 - Check Diagnosis Compatibility
Necessity Analysis Tab	Access necessity analysis functionality	User click input	Core component for UC_38 analysis

Name of item	**Description of purpose**	**Source of input or destination of output**	**Relationships to other inputs/outputs**
Recommendations Tab	Access compatibility recommendations	User click input	Output component for UC_38 analysis
Performing Necessity Analysis Alert	Confirm necessity analysis is in progress	Analysis system status	Indicates UC_38 process execution
Treatment Plan Analysis Information	Show patient and analysis context	Treatment plan database	Provides UC_38 analysis context
Treatment Plan Necessity Score	Display overall necessity percentage	Necessity analysis algorithm	Core UC_38 analysis result
Necessity Score Display	Show percentage necessity score	Necessity calculation system	Primary UC_38 output metric
Save Analysis Button	Store compatibility analysis results	User click input	UC_38 analysis preservation
Generate Report Button	Create compatibility analysis report	User click input	UC_38 report generation
Mark as Reviewed Button	Indicate compatibility review completion	User click input	UC_38 review completion action

